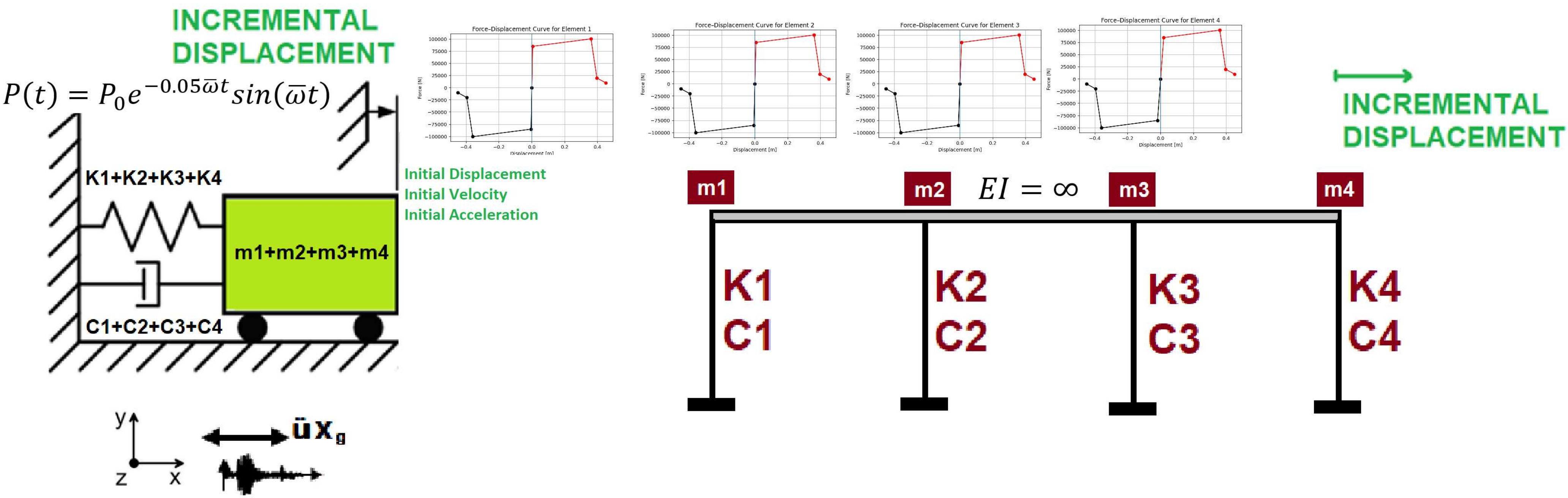
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

SENSITIVITY ANALYSIS OF SINGLE-DEGREE-FREEOM (SDOF) STRUCTURES USING FREE-VIBRATION EFFECTS OF INTIAL DISPLACEMENT, STRUCTURAL ELASTIC STIFFNESS, STRUCTURAL DUCTILITY RATIO AND OVER-STRENGTH FACTOR ON OUTPUT KEY PARAMETERS SUCH AS EQULIVALENT VISCOUS DAMPING RATIO FROM NONLINEAR DYNAMIC ANALYSES USING PYTHON AND OPENSEES

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)


INCREMENTAL DISPLACEMENT
$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$
K1+K2+K3+K4
m1+m2+m3+m4
C1+C2+C3+C4
Initial Displacement
Initial Velocity
Initial Acceleration
Force-Displacement Curve for Element 1
Force-Displacement Curve for Element 2
Force-Displacement Curve for Element 3
Force-Displacement Curve for Element 4
Displacement [m]
Force [N]
INCREMENTAL DISPLACEMENT
m1
m2
m3
m4
$EI = \infty$
K1
C1
K2
C2
K3
C3
K4
C4
y
x
z
$\ddot{u}x_g$

$$Structural\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

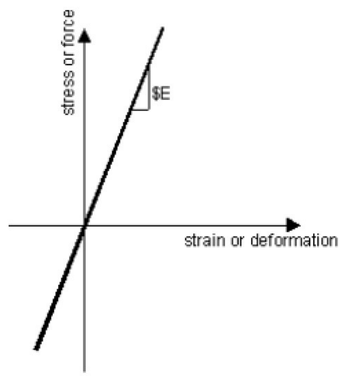
$$\Delta_d = Lateral\ Displaement\ from\ Dynamic\ Analysis$$

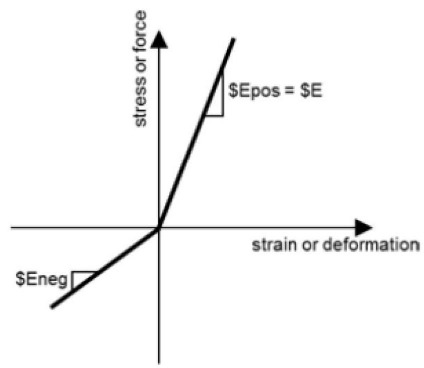
$$\Delta_y = Lateral\ Yield\ Displaement\ from\ Pushover\ Analysis$$

$$\Delta_u = Lateral\ Ultimate\ Displaement\ from\ Pushover\ Analysis$$

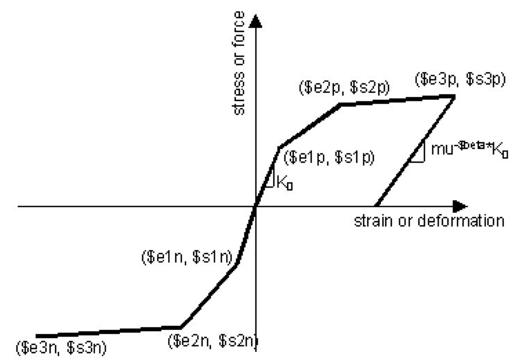
$$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$





ELASTIC MATERIAL



INELASTIC MATERIAL

Energy quantities and the damping formula

- **Dissipated energy** E_d – the area enclosed by the hysteresis loop. This is the net work done per cycle and represents the energy that must be removed from the system to return it to the original state. Physically, it corresponds to plastic yielding, friction, cracking, or other irreversible mechanisms.
- **Maximum strain energy** E_s – defined here as

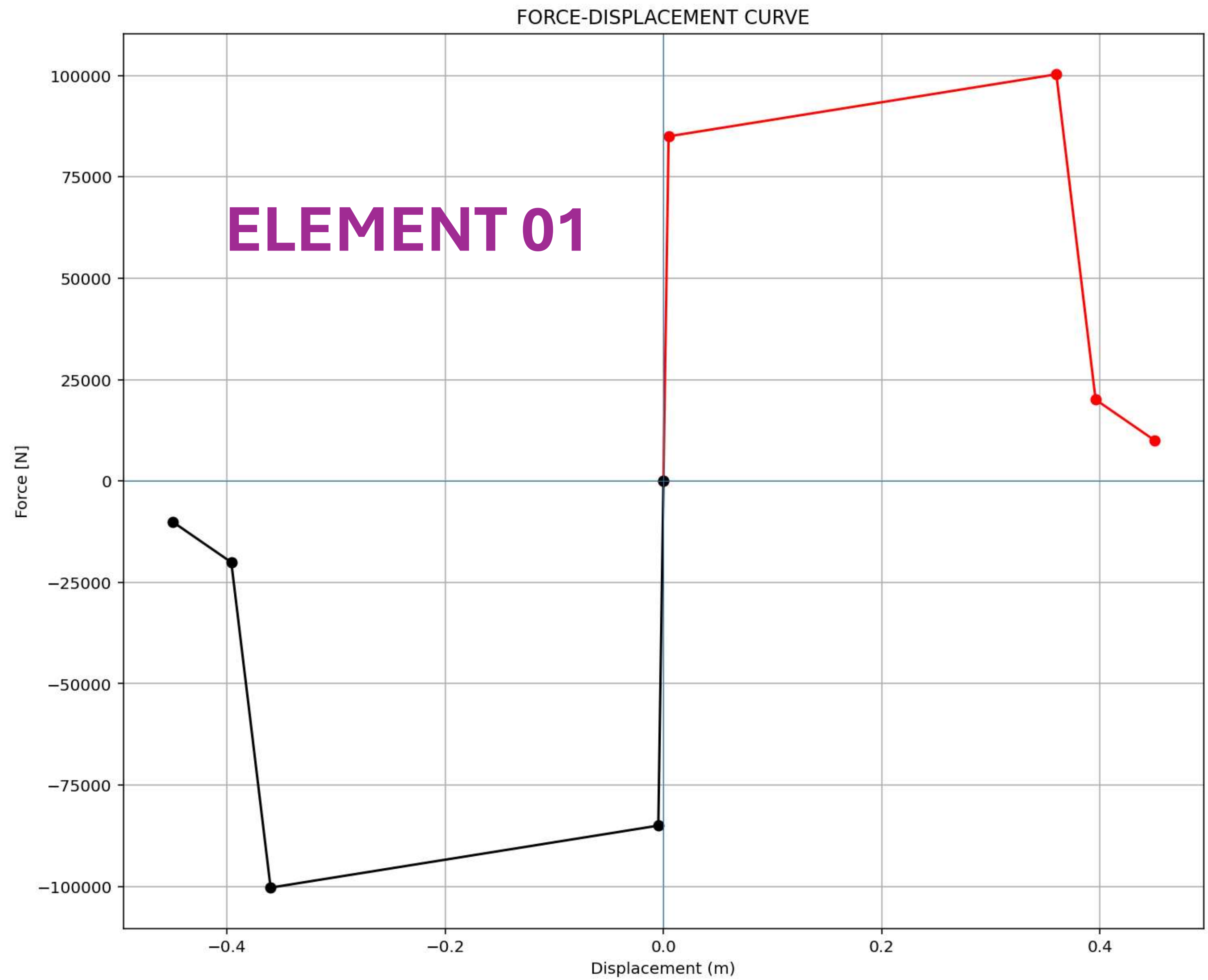
$$E_s = \frac{1}{2} F_{\max} u_{\max}$$

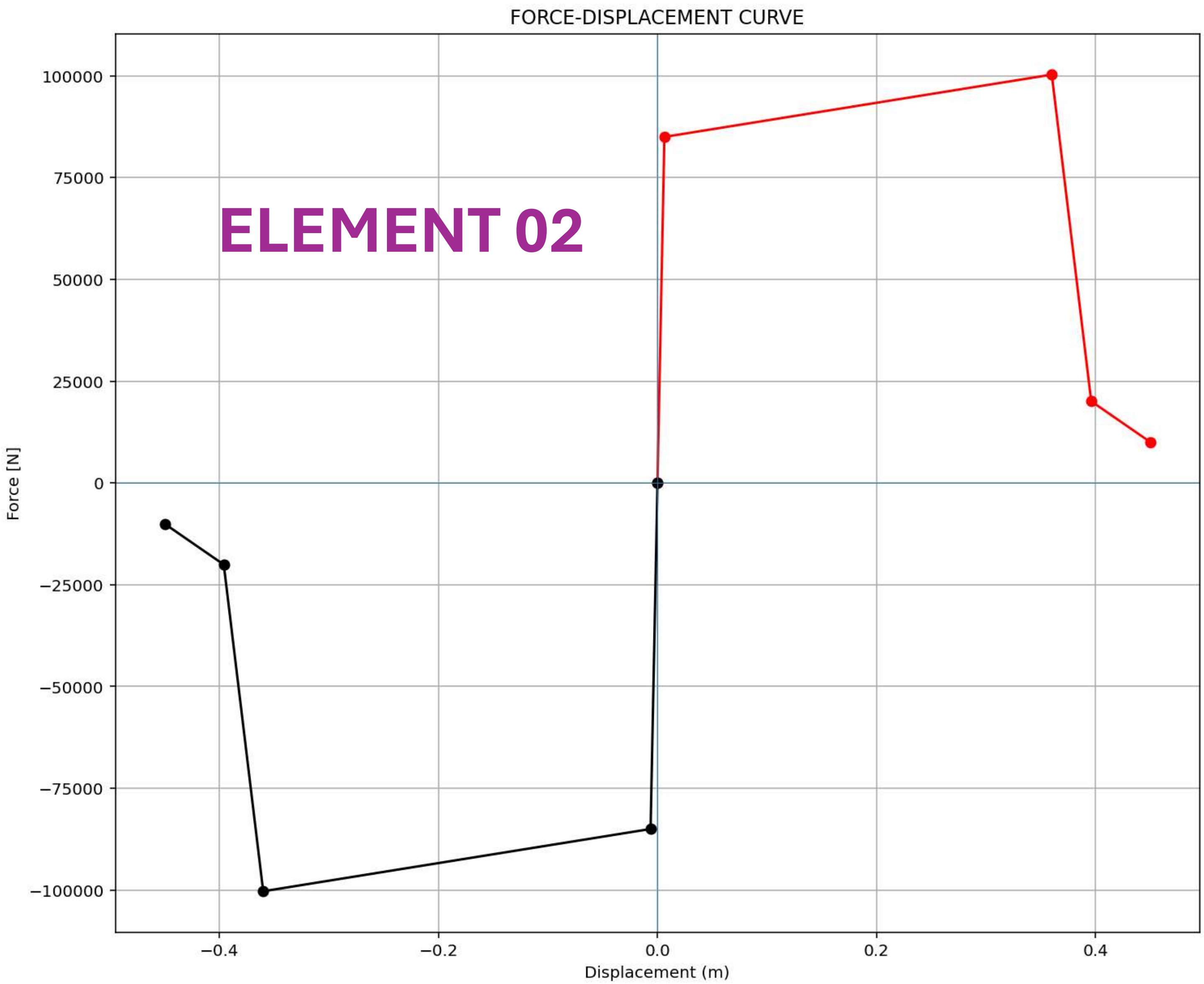
where $u_{\max} = \max |u|$ is the peak absolute displacement in the loop and $F_{\max}$ is the absolute value of the force **at that same displacement**. This definition assumes a **secant stiffness** $k_{sec} = F_{\max}/u_{\max}$ and represents the elastic strain energy stored at peak displacement if the system were linear with that stiffness.

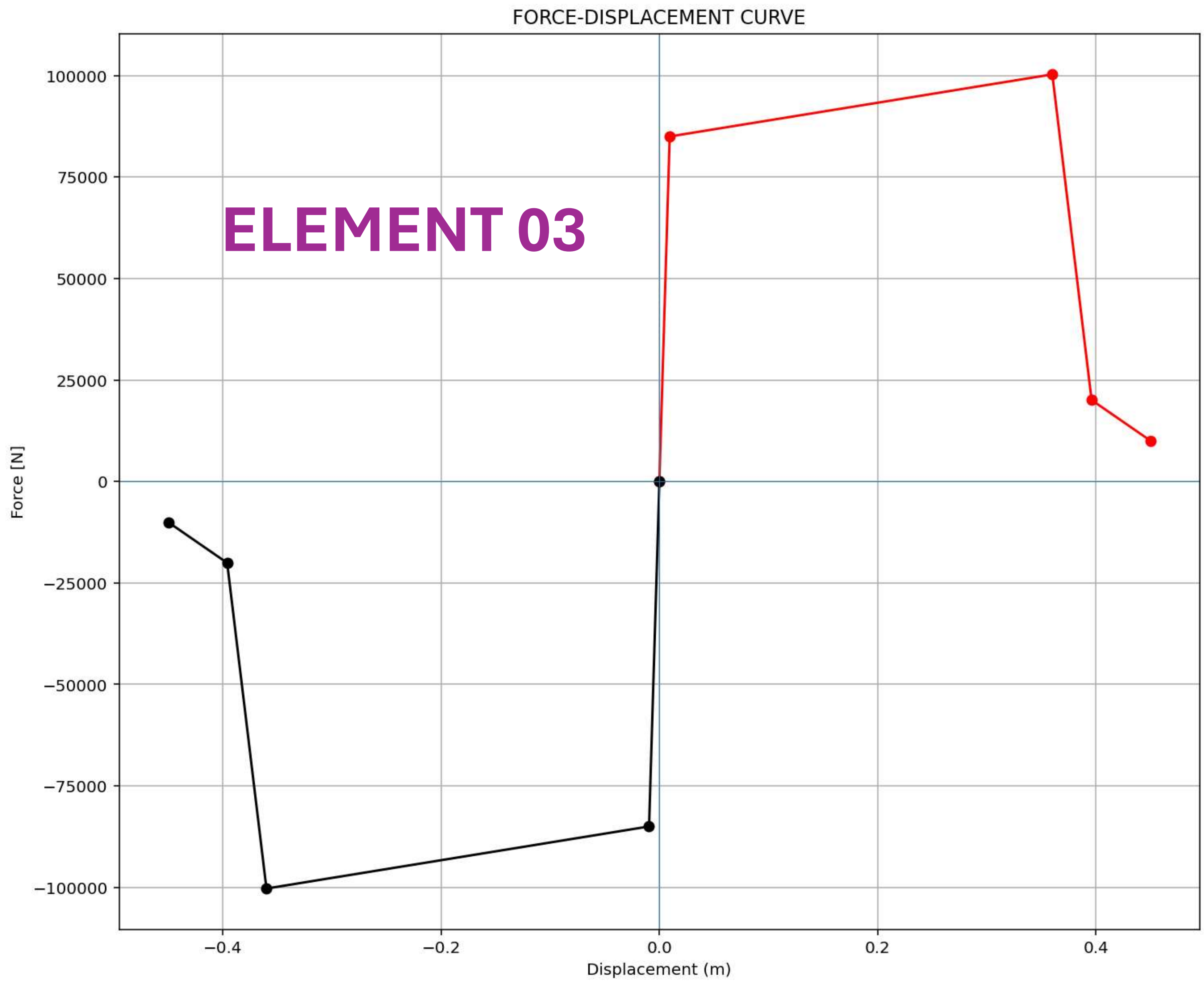
The equivalent viscous damping ratio is then:

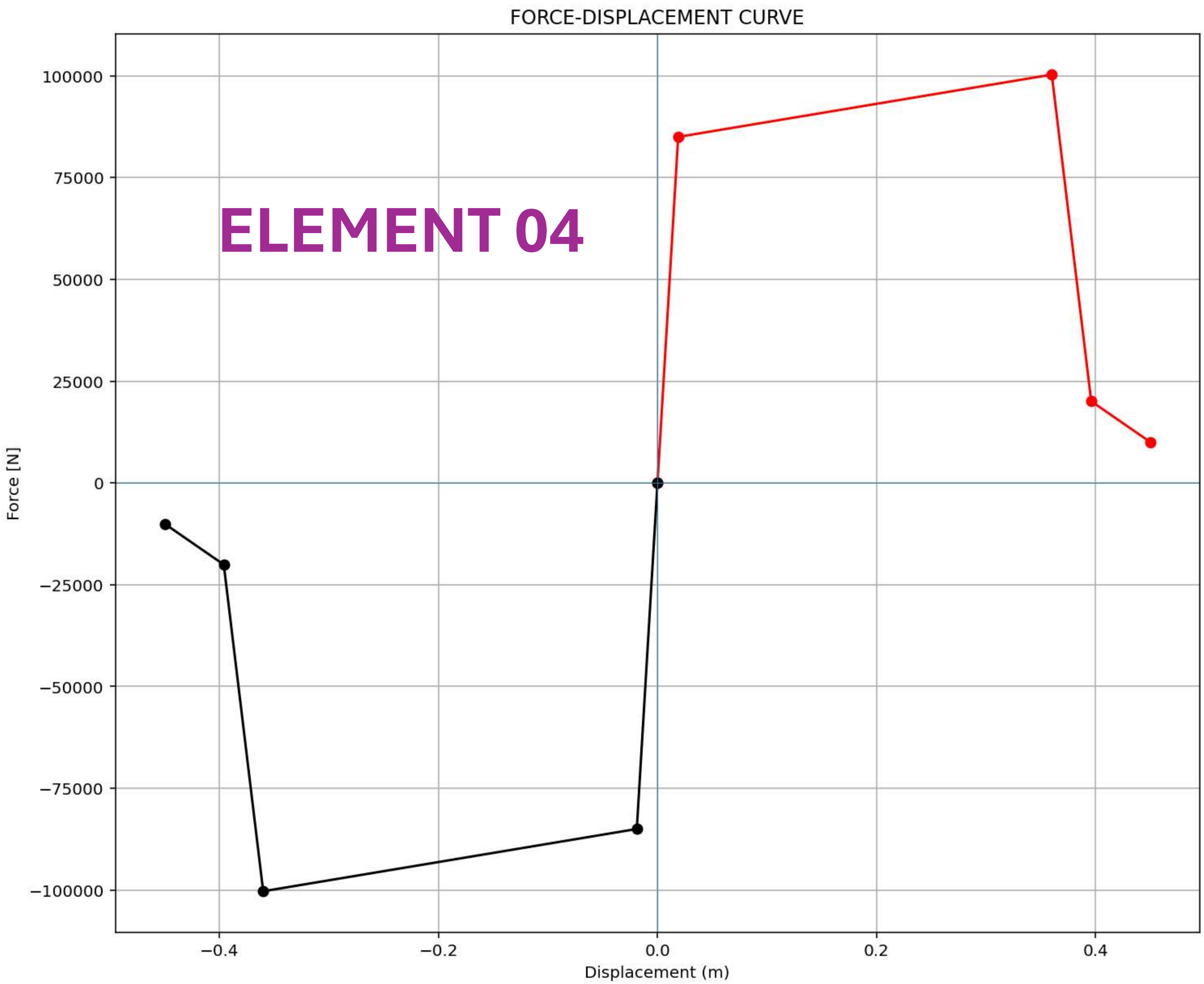
$$\xi_{eq} = \frac{100\, E_d}{4\pi\, E_s} \quad [\%]$$

This stems from equating E_d to the energy dissipated per cycle by a linear viscous damper, $E_d = \pi c \omega u_{\max}^2$, while $E_s = \frac{1}{2} k u_{\max}^2$; substituting $\xi = c/(2m\omega)$ and $\omega^2 = k/m$ yields the familiar expression $\xi = E_d/(4\pi E_s)$.

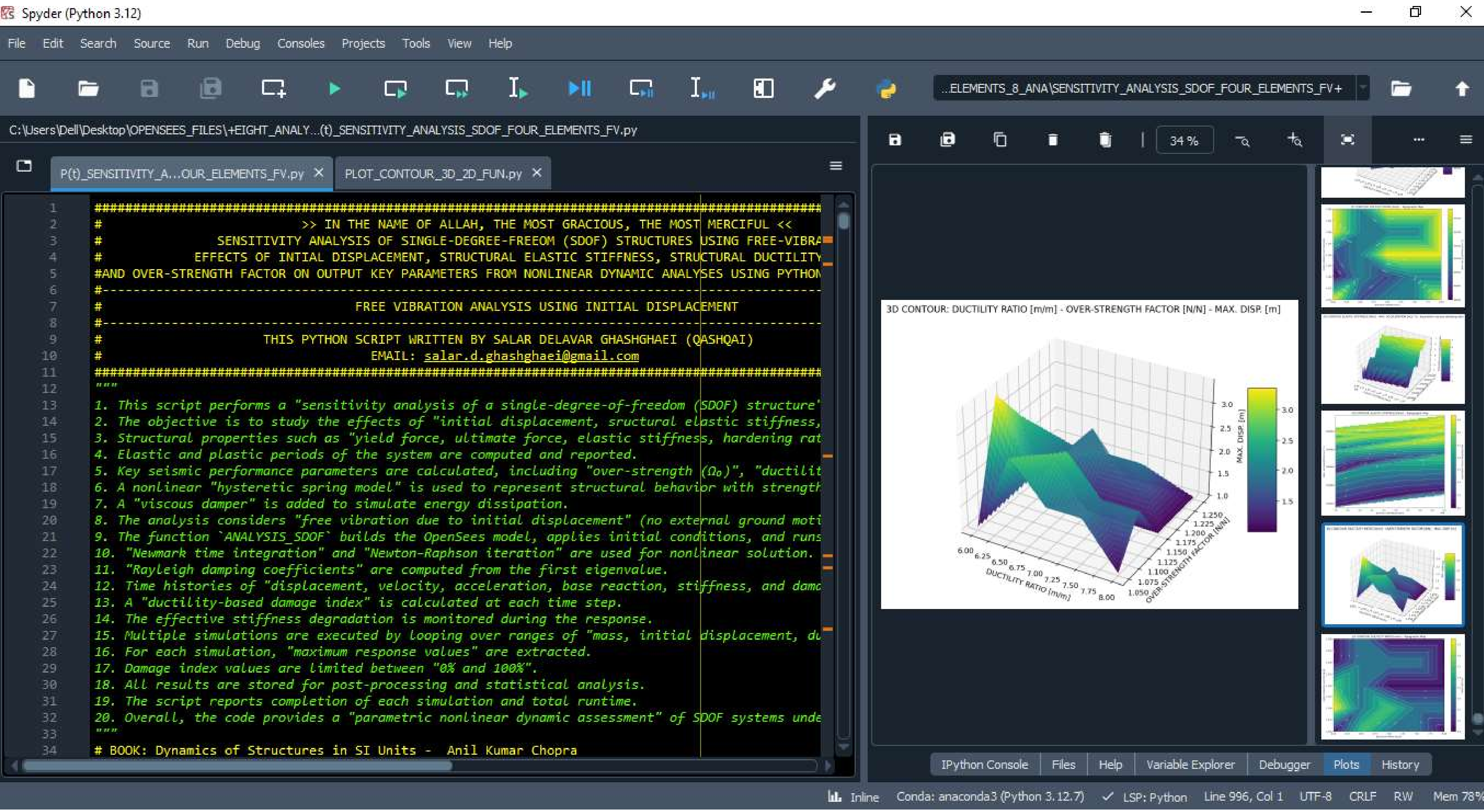

FORCE-DISPLACEMENT CURVE
ELEMENT 01
Force [N]
Displacement (m)
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
−0.4
−0.2
0.0
0.2
0.4


FORCE-DISPLACEMENT CURVE
ELEMENT 02
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
Force [N]
−0.4
−0.2
0.0
0.2
0.4
Displacement (m)


FORCE-DISPLACEMENT CURVE
ELEMENT 03
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
Force [N]
−0.4
−0.2
0.0
0.2
0.4
Displacement (m)


FORCE-DISPLACEMENT CURVE
ELEMENT 04
Force [N]
Displacement (m)
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
−0.4
−0.2
0.0
0.2
0.4

FREE-VIBRATION ANALYSIS



Spyder (Python 3.12)
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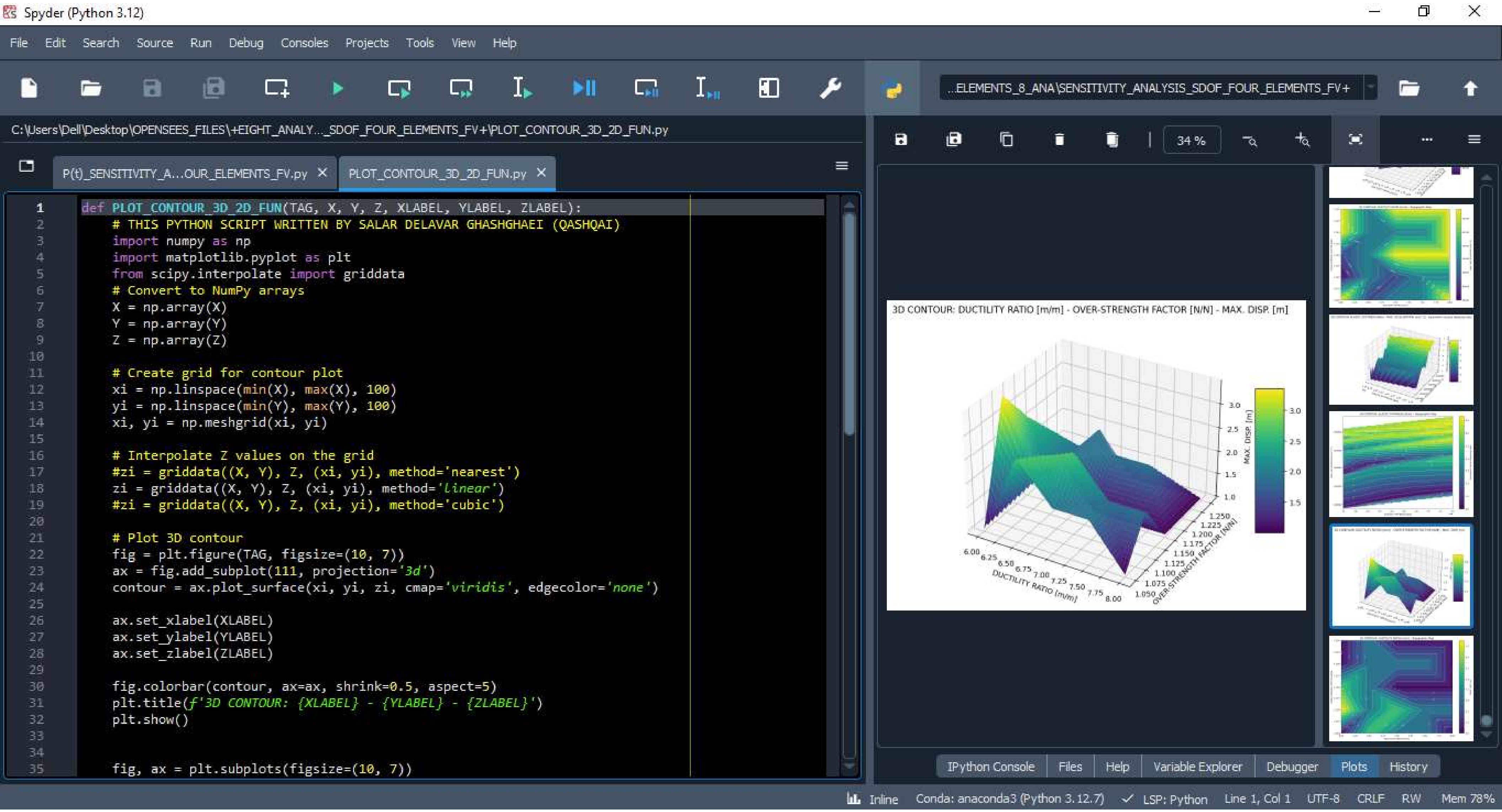
P(t)_SENSITIVITY_A...OUR_ELEMENTS_FV.py
PLOT_CONTOUR_3D_2D_FUN.py
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
SENSITIVITY ANALYSIS OF SINGLE-DEGREE-FREEOM (SDOF) STRUCTURES USING FREE-VIBRA
EFFECTS OF INTIAL DISPLACEMENT, STRUCTURAL ELASTIC STIFFNESS, STRUCTURAL DUCTILITY
#AND OVER-STRENGTH FACTOR ON OUTPUT KEY PARAMETERS FROM NONLINEAR DYNAMIC ANALYSES USING PYTHON
FREE VIBRATION ANALYSIS USING INITIAL DISPLACEMENT
THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
EMAIL: salar.d.ghashghaei@gmail.com
"""
1. This script performs a "sensitivity analysis of a single-degree-of-freedom (SDOF) structure"
2. The objective is to study the effects of "initial displacement, sructural elastic stiffness,
3. Structural properties such as "yield force, ultimate force, elastic stiffness, hardening rat
4. Elastic and plastic periods of the system are computed and reported.
5. Key seismic performance parameters are calculated, including "over-strength (Ω₀)", "ductilit
6. A nonlinear "hysteretic spring model" is used to represent structural behavior with strength
7. A "viscous damper" is added to simulate energy dissipation.
8. The analysis considers "free vibration due to initial displacement" (no external ground moti
9. The function `ANALYSIS_SDOF` builds the OpenSees model, applies initial conditions, and runs
10. "Newmark time integration" and "Newton-Raphson iteration" are used for nonlinear solution.
11. "Rayleigh damping coefficients" are computed from the first eigenvalue.
12. Time histories of "displacement, velocity, acceleration, base reaction, stiffness, and damo
13. A "ductility-based damage index" is calculated at each time step.
14. The effective stiffness degradation is monitored during the response.
15. Multiple simulations are executed by looping over ranges of "mass, initial displacement, du
16. For each simulation, "maximum response values" are extracted.
17. Damage index values are limited between "0% and 100%".
18. All results are stored for post-processing and statistical analysis.
19. The script reports completion of each simulation and total runtime.
20. Overall, the code provides a "parametric nonlinear dynamic assessment" of SDOF systems unde
"""
BOOK: Dynamics of Structures in SI Units - Anil Kumar Chopra
34 %
3D CONTOUR: DUCTILITY RATIO [m/m] - OVER-STRENGTH FACTOR [N/N] - MAX. DISP. [m]
DUCTILITY RATIO [m/m]
OVER-STRENGTH FACTOR [N/N]
MAX. DISP. [m]
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 996, Col 1 UTF-8 CRLF RW Mem 78%


Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
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C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY..._SDOF_FOUR_ELEMENTS_FV+\PLOT_CONTOUR_3D_2D_FUN.py
P(t)_SENSITIVITY_A...OUR_ELEMENTS_FV.py
PLOT_CONTOUR_3D_2D_FUN.py
def PLOT_CONTOUR_3D_2D_FUN(TAG, X, Y, Z, XLABEL, YLABEL, ZLABEL):
    # THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
    import numpy as np
    import matplotlib.pyplot as plt
    from scipy.interpolate import griddata
    # Convert to NumPy arrays
    X = np.array(X)
    Y = np.array(Y)
    Z = np.array(Z)

    # Create grid for contour plot
    xi = np.linspace(min(X), max(X), 100)
    yi = np.linspace(min(Y), max(Y), 100)
    xi, yi = np.meshgrid(xi, yi)

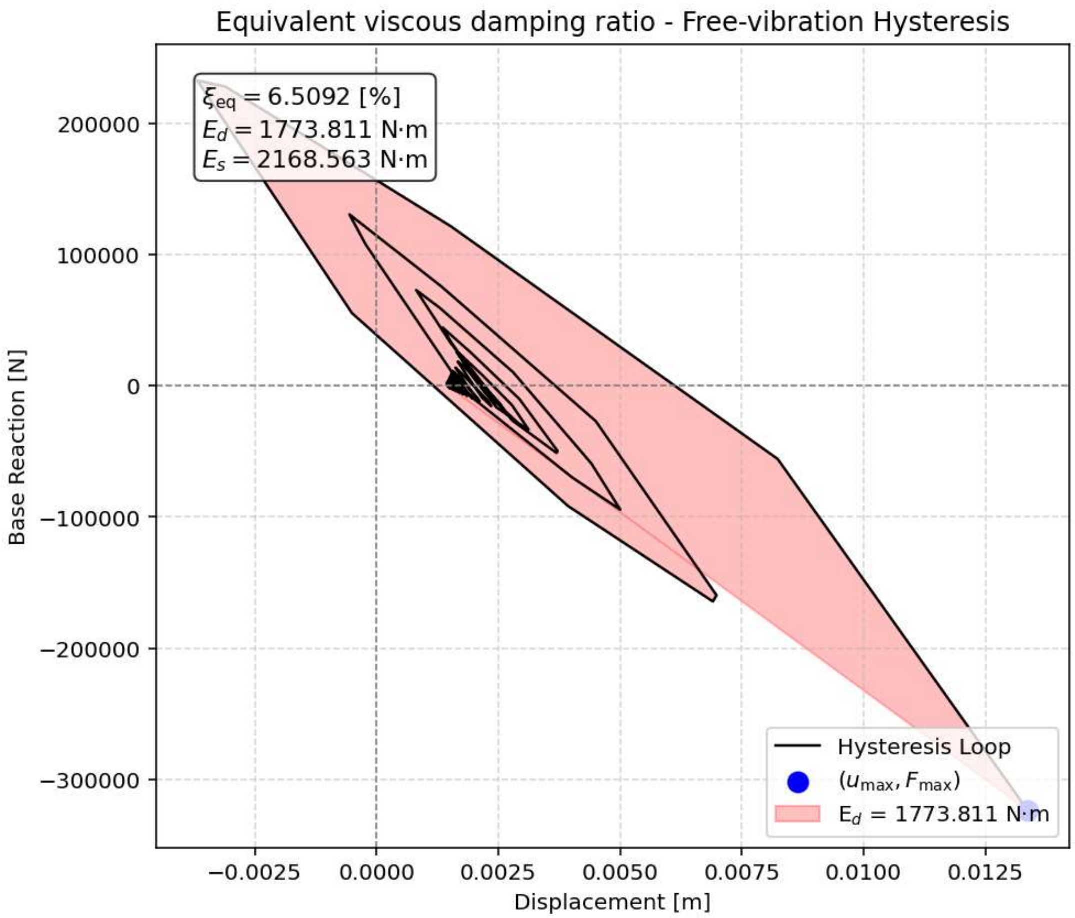
    # Interpolate Z values on the grid
    #zi = griddata((X, Y), Z, (xi, yi), method='nearest')
    zi = griddata((X, Y), Z, (xi, yi), method='linear')
    #zi = griddata((X, Y), Z, (xi, yi), method='cubic')

    # Plot 3D contour
    fig = plt.figure(TAG, figsize=(10, 7))
    ax = fig.add_subplot(111, projection='3d')
    contour = ax.plot_surface(xi, yi, zi, cmap='viridis', edgecolor='none')

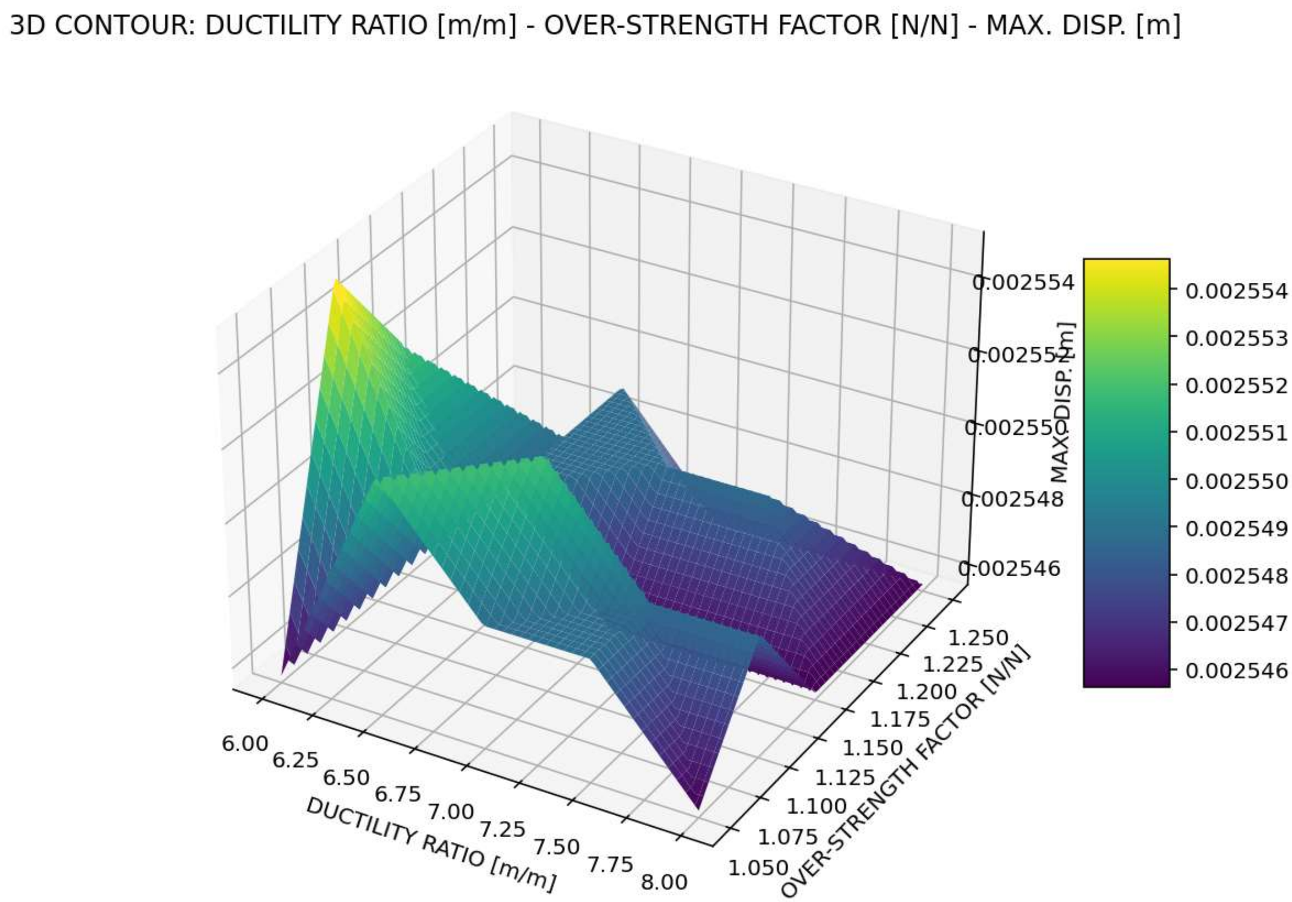
    ax.set_xlabel(XLABEL)
    ax.set_ylabel(YLABEL)
    ax.set_zlabel(ZLABEL)

    fig.colorbar(contour, ax=ax, shrink=0.5, aspect=5)
    plt.title(f'3D CONTOUR: {XLABEL} - {YLABEL} - {ZLABEL}')
    plt.show()

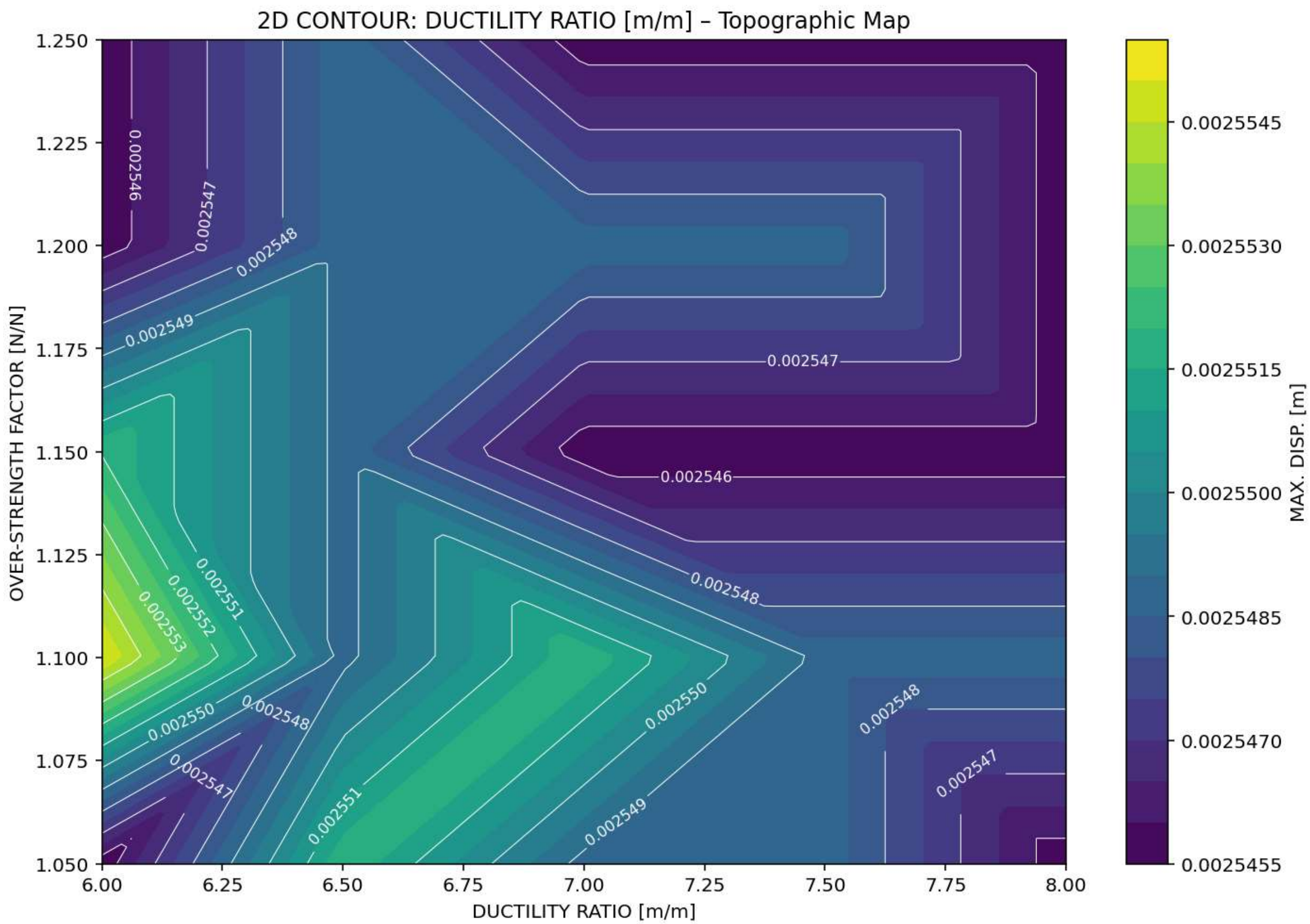
    fig, ax = plt.subplots(figsize=(10, 7))
34 %
3D CONTOUR: DUCTILITY RATIO [m/m] - OVER-STRENGTH FACTOR [N/N] - MAX. DISP. [m]
DUCTILITY RATIO [m/m]
OVER-STRENGTH FACTOR [N/N]
MAX. DISP. [m]
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 78%



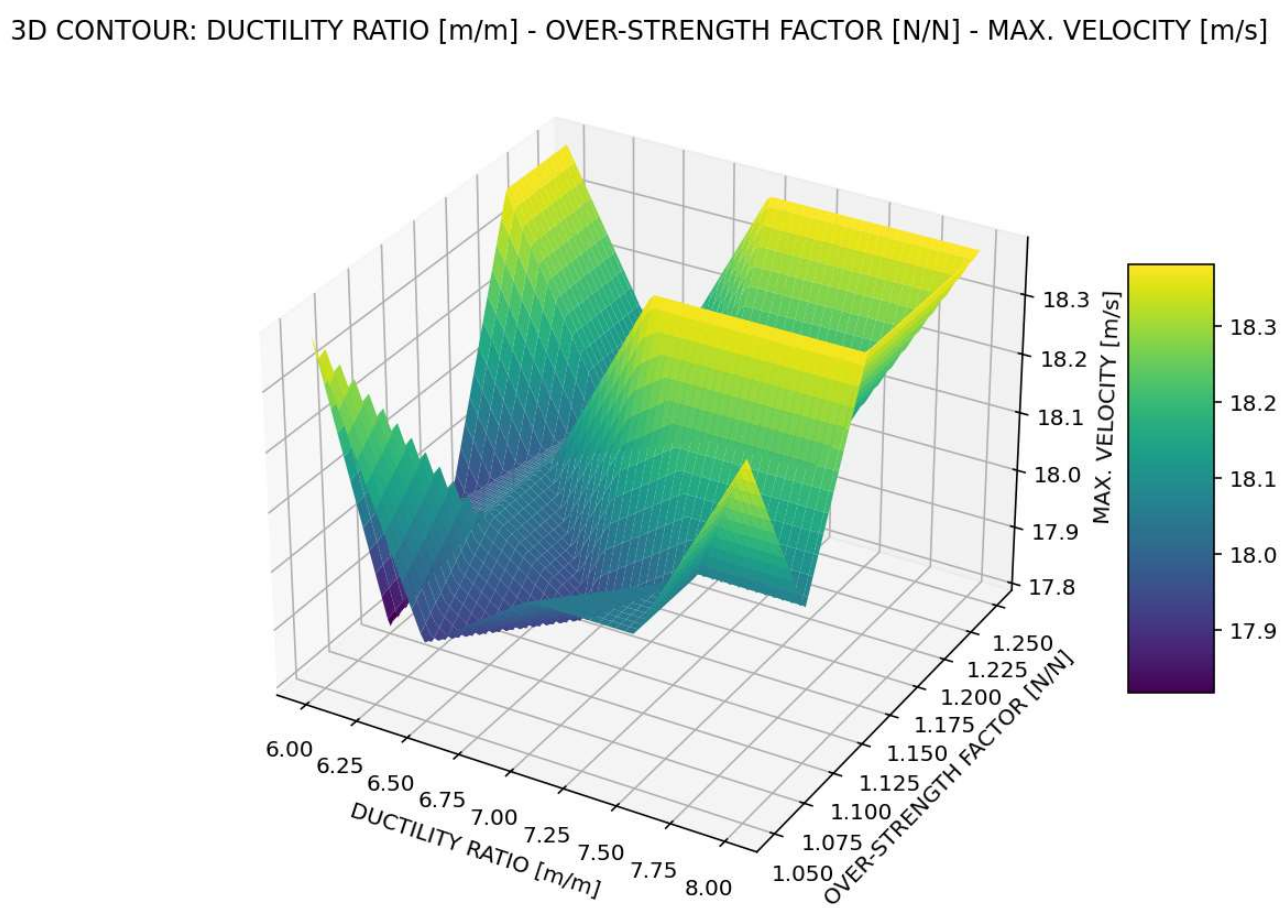
Equivalent viscous damping ratio - Free-vibration Hysteresis
$\xi_{eq} = 6.5092$ [%]
$E_d = 1773.811$ N·m
$E_s = 2168.563$ N·m
Hysteresis Loop
(u_{max}, F_{max})
E_d = 1773.811 N·m
Base Reaction [N]
Displacement [m]
200000
100000
0
−100000
−200000
−300000
−0.0025
0.0000
0.0025
0.0050
0.0075
0.0100
0.0125



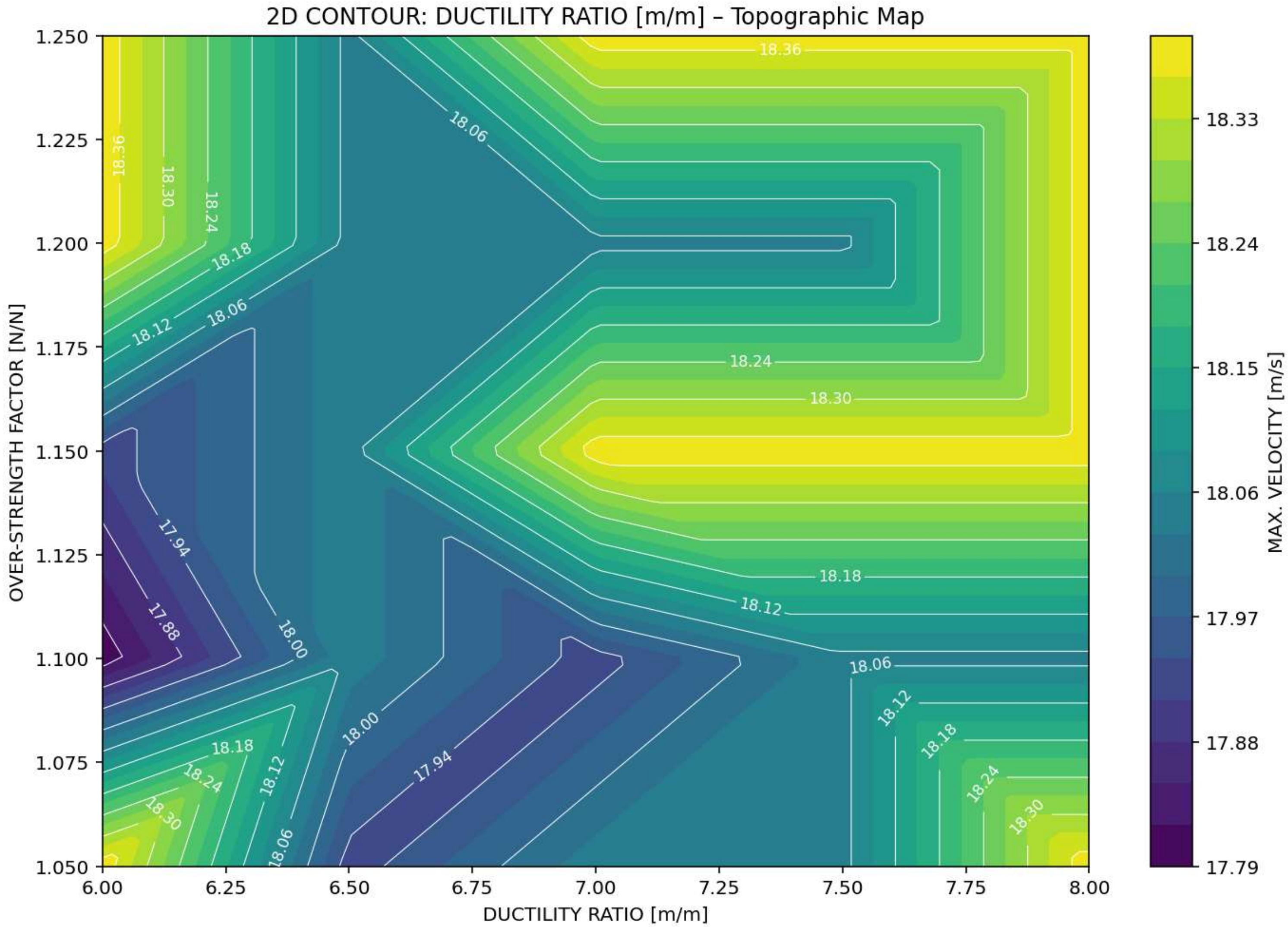
3D CONTOUR: DUCTILITY RATIO [m/m] - OVER-STRENGTH FACTOR [N/N] - MAX. DISP. [m]
DUCTILITY RATIO [m/m]
6.00
6.25
6.50
6.75
7.00
7.25
7.50
7.75
8.00
OVER-STRENGTH FACTOR [N/N]
1.050
1.075
1.100
1.125
1.150
1.175
1.200
1.225
1.250
MAX. DISP. [m]
0.002546
0.002548
0.002554
0.002546
0.002547
0.002548
0.002549
0.002550
0.002551
0.002552
0.002553
0.002554



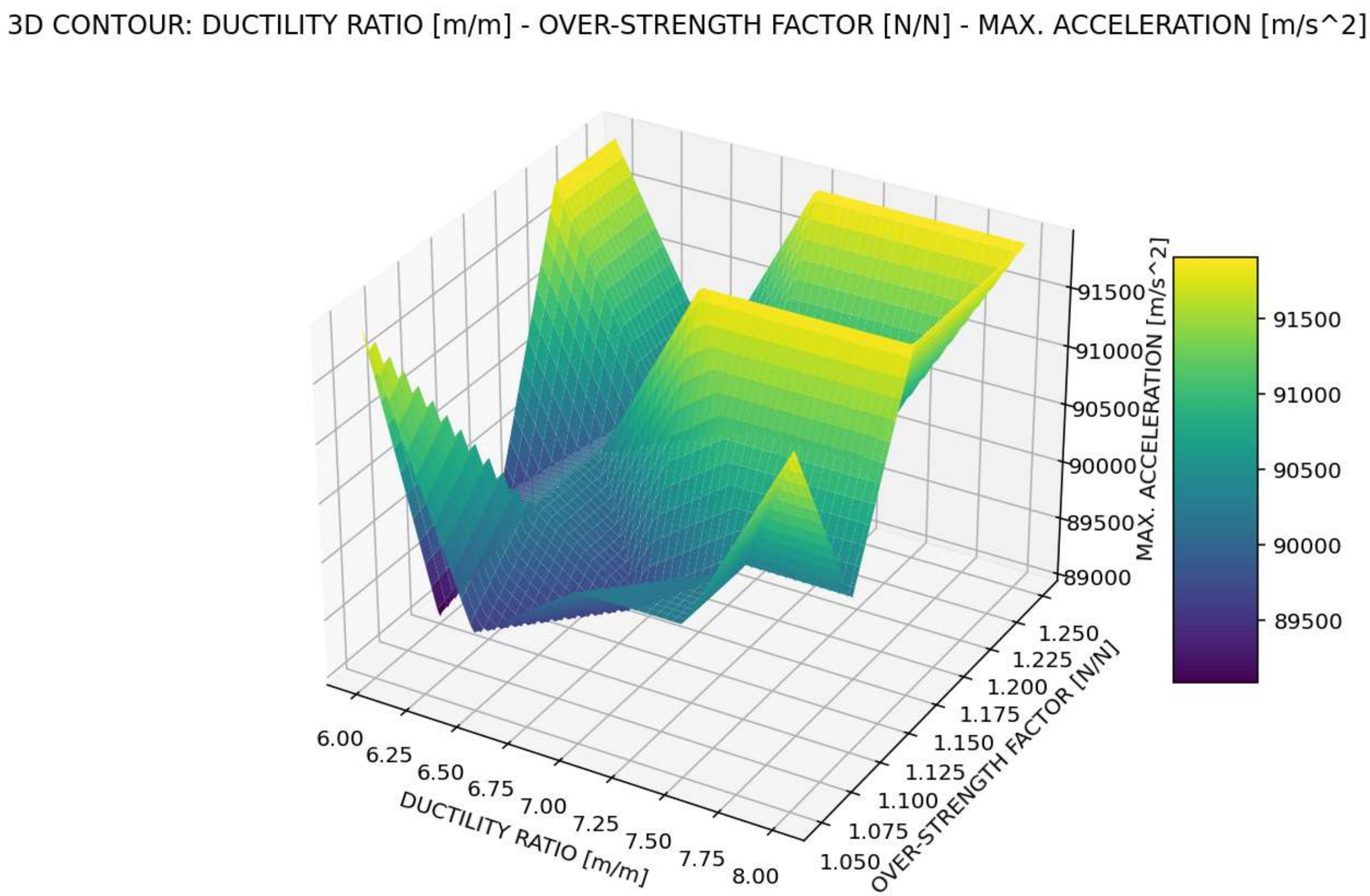
2D CONTOUR: DUCTILITY RATIO [m/m] – Topographic Map
OVER-STRENGTH FACTOR [N/N]
DUCTILITY RATIO [m/m]
MAX. DISP. [m]
1.250
1.225
1.200
1.175
1.150
1.125
1.100
1.075
1.050
6.00
6.25
6.50
6.75
7.00
7.25
7.50
7.75
8.00
0.0025545
0.0025530
0.0025515
0.0025500
0.0025485
0.0025470
0.0025455
0.002546
0.002547
0.002548
0.002549
0.002550
0.002551
0.002552
0.002553

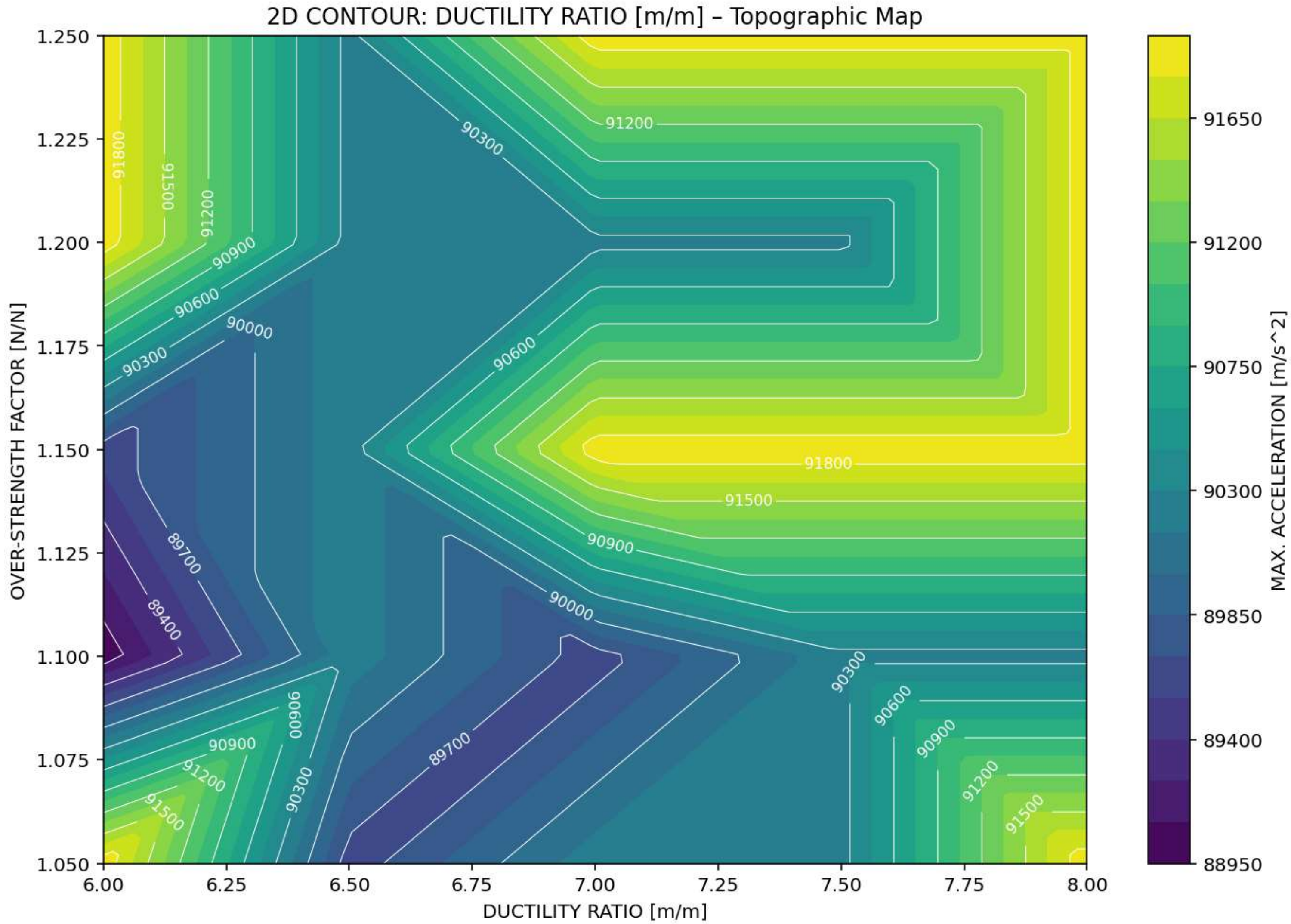


3D CONTOUR: DUCTILITY RATIO [m/m] - OVER-STRENGTH FACTOR [N/N] - MAX. VELOCITY [m/s]
DUCTILITY RATIO [m/m]
6.00
6.25
6.50
6.75
7.00
7.25
7.50
7.75
8.00
OVER-STRENGTH FACTOR [N/N]
1.050
1.075
1.100
1.125
1.150
1.175
1.200
1.225
1.250
MAX. VELOCITY [m/s]
17.8
17.9
18.0
18.1
18.2
18.3



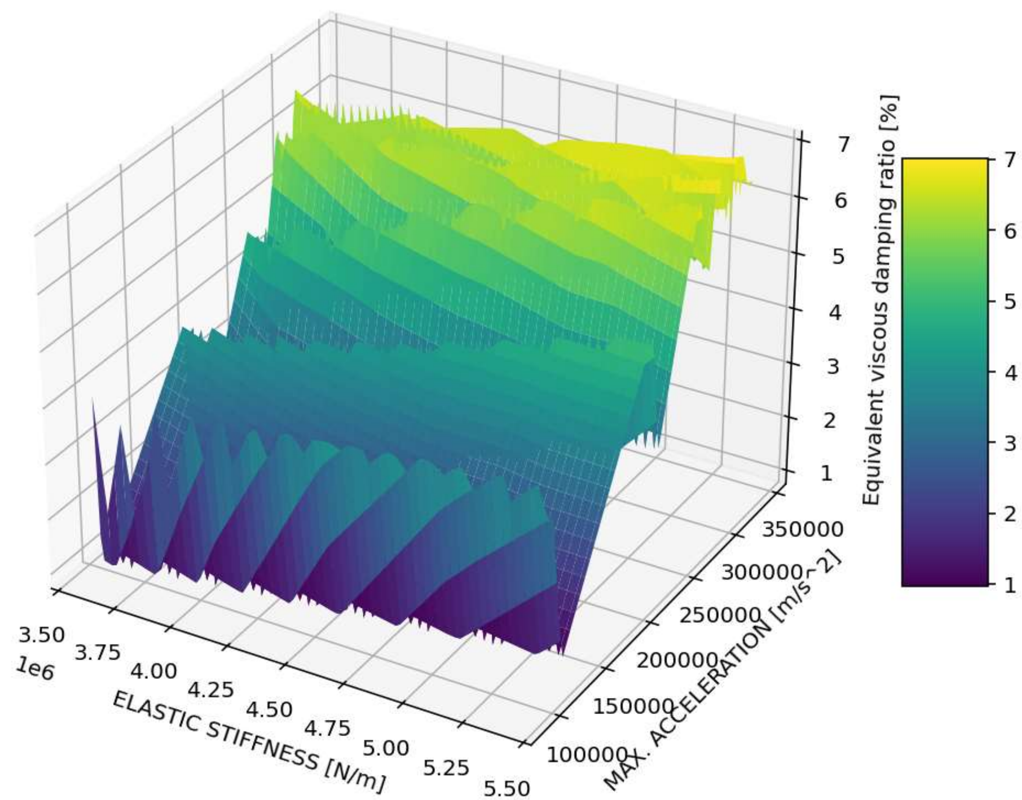
2D CONTOUR: DUCTILITY RATIO [m/m] – Topographic Map
OVER-STRENGTH FACTOR [N/N]
DUCTILITY RATIO [m/m]
MAX. VELOCITY [m/s]
1.250
1.225
1.200
1.175
1.150
1.125
1.100
1.075
1.050
6.00
6.25
6.50
6.75
7.00
7.25
7.50
7.75
8.00
18.33
18.24
18.15
18.06
17.97
17.88
17.79
18.36
18.30
18.24
18.18
18.12
18.06
18.00
17.94
17.88

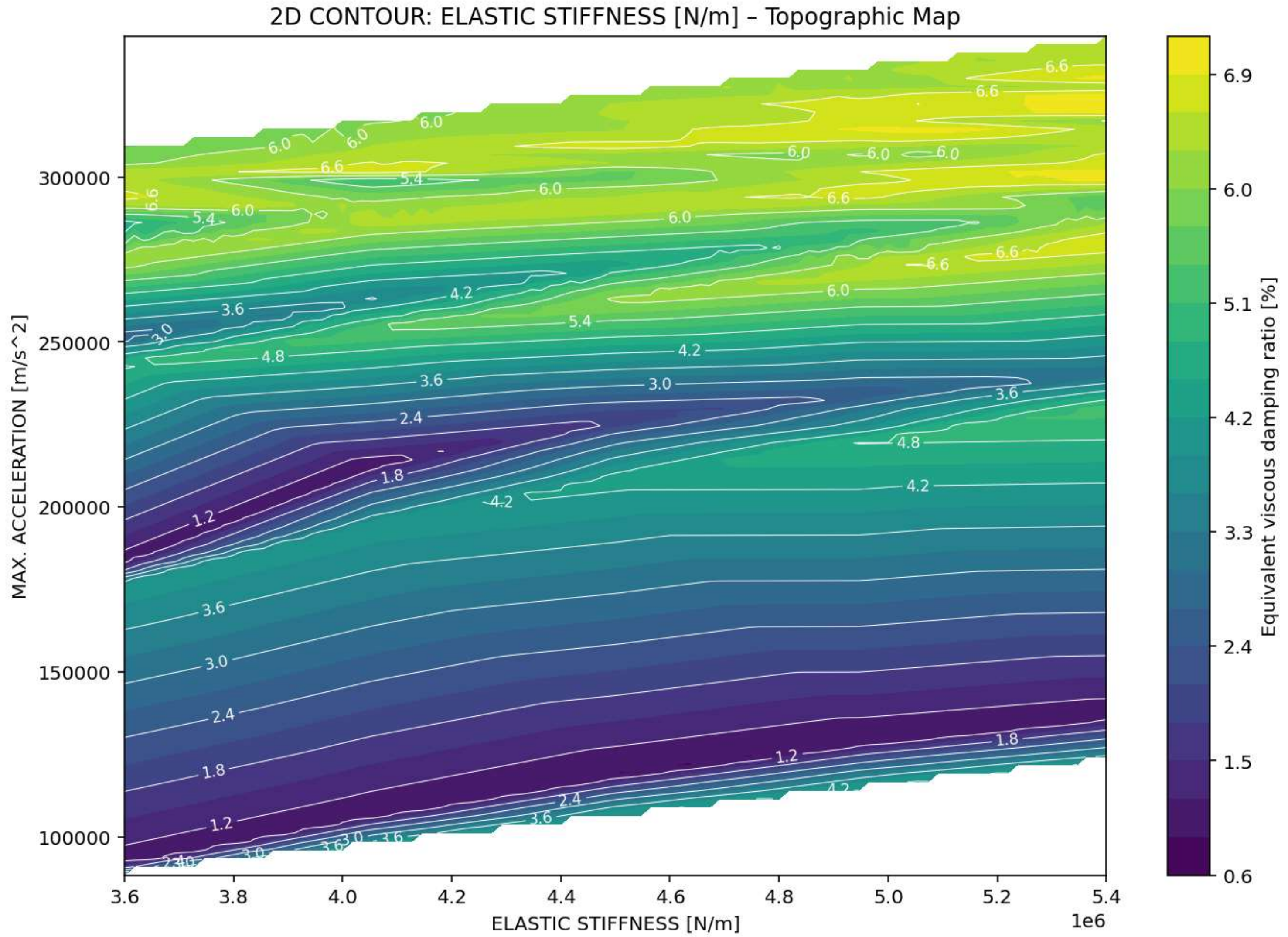

3D CONTOUR: DUCTILITY RATIO [m/m] - OVER-STRENGTH FACTOR [N/N] - MAX. ACCELERATION [m/s^2]
DUCTILITY RATIO [m/m]
6.00
6.25
6.50
6.75
7.00
7.25
7.50
7.75
8.00
OVER-STRENGTH FACTOR [N/N]
1.050
1.075
1.100
1.125
1.150
1.175
1.200
1.225
1.250
MAX. ACCELERATION [m/s^2]
89000
89500
90000
90500
91000
91500

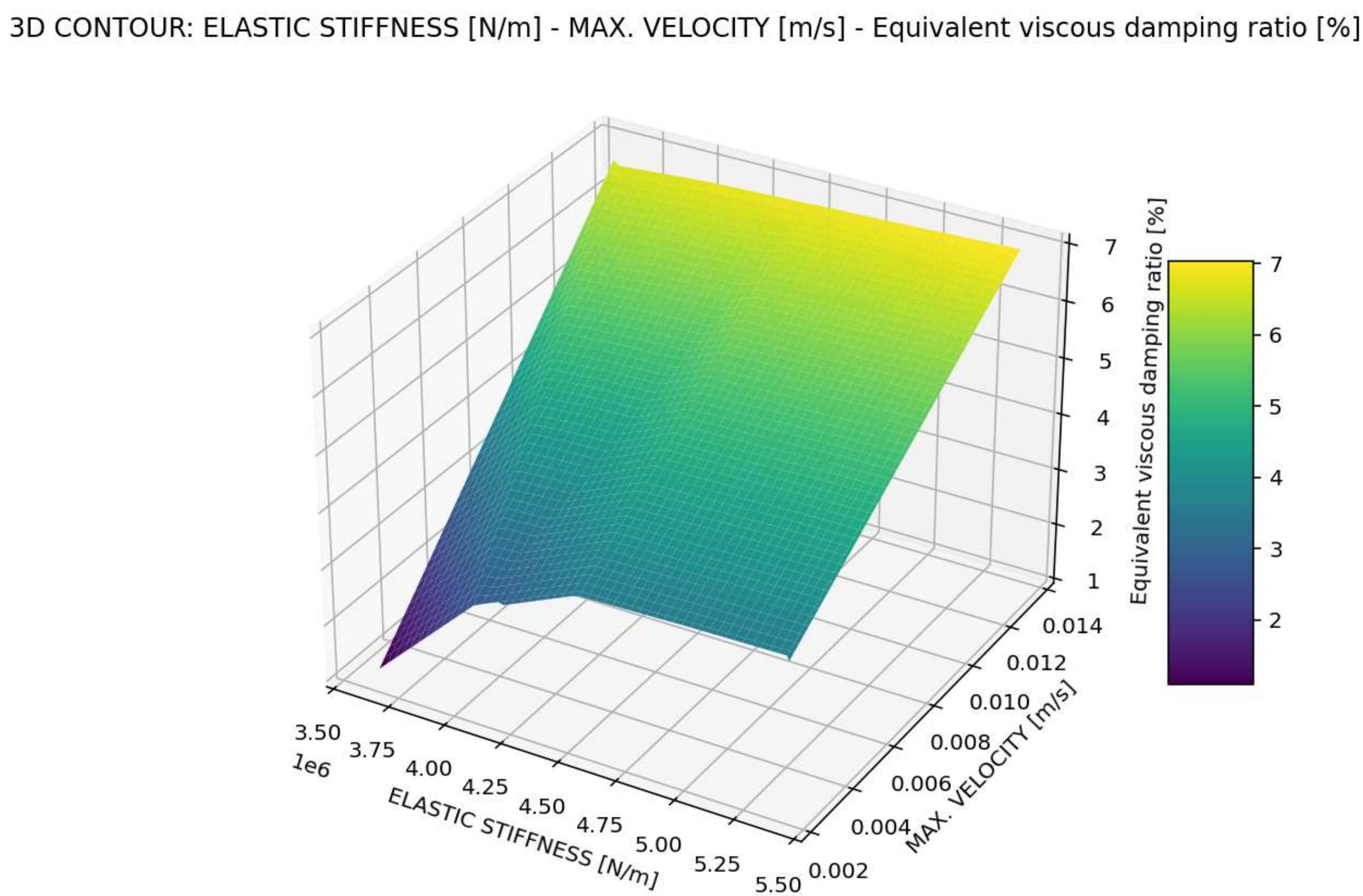


2D CONTOUR: DUCTILITY RATIO [m/m] – Topographic Map
OVER-STRENGTH FACTOR [N/N]
DUCTILITY RATIO [m/m]
MAX. ACCELERATION [m/s^2]
1.250
1.225
1.200
1.175
1.150
1.125
1.100
1.075
1.050
6.00
6.25
6.50
6.75
7.00
7.25
7.50
7.75
8.00
91650
91200
90750
90300
89850
89400
88950
91800
91500
91200
90900
90600
90300
90000
89700
89400

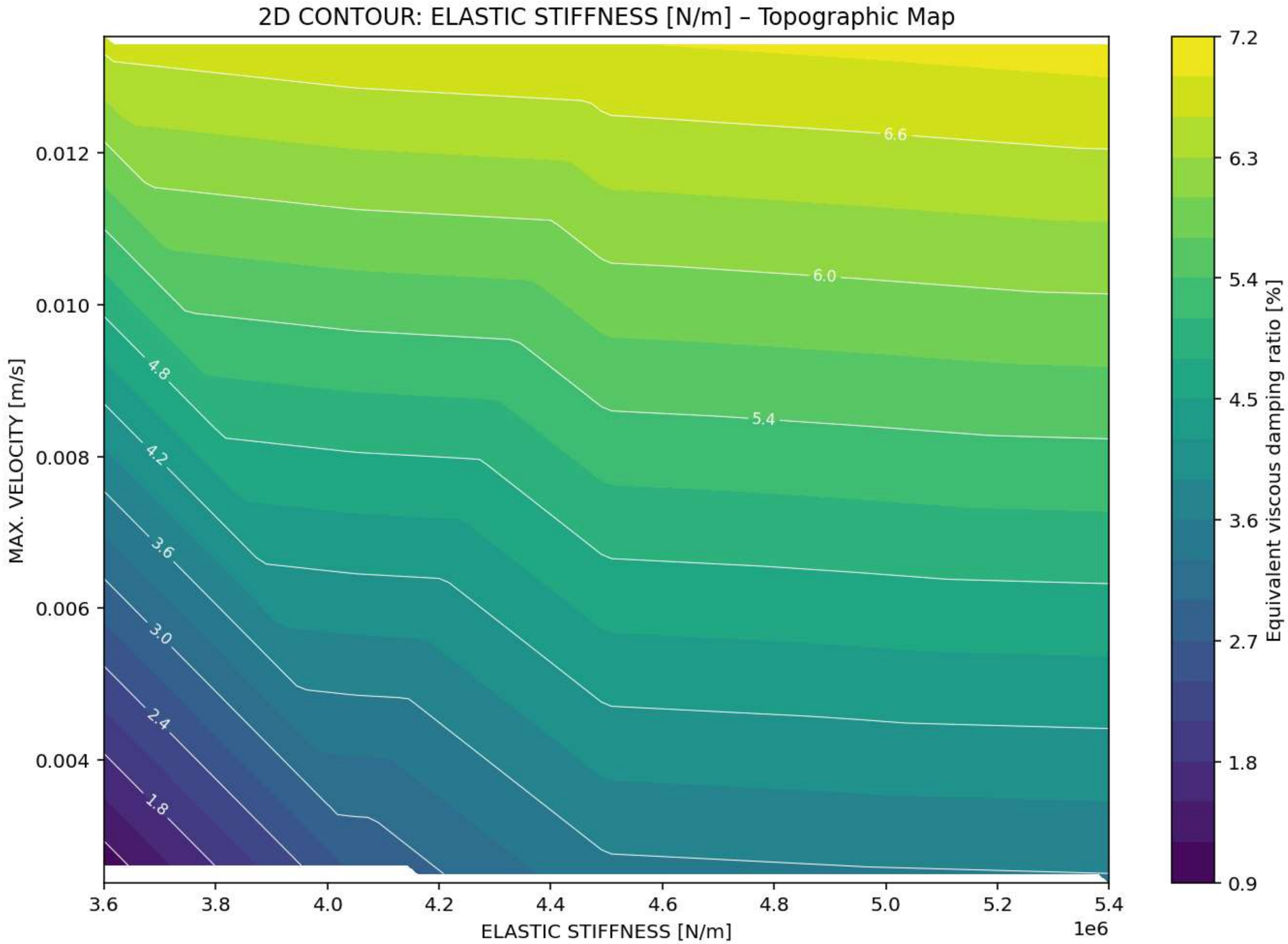
3D CONTOUR: ELASTIC STIFFNESS [N/m] - MAX. ACCELERATION [m/s^2] - Equivalent viscous damping ratio [%]

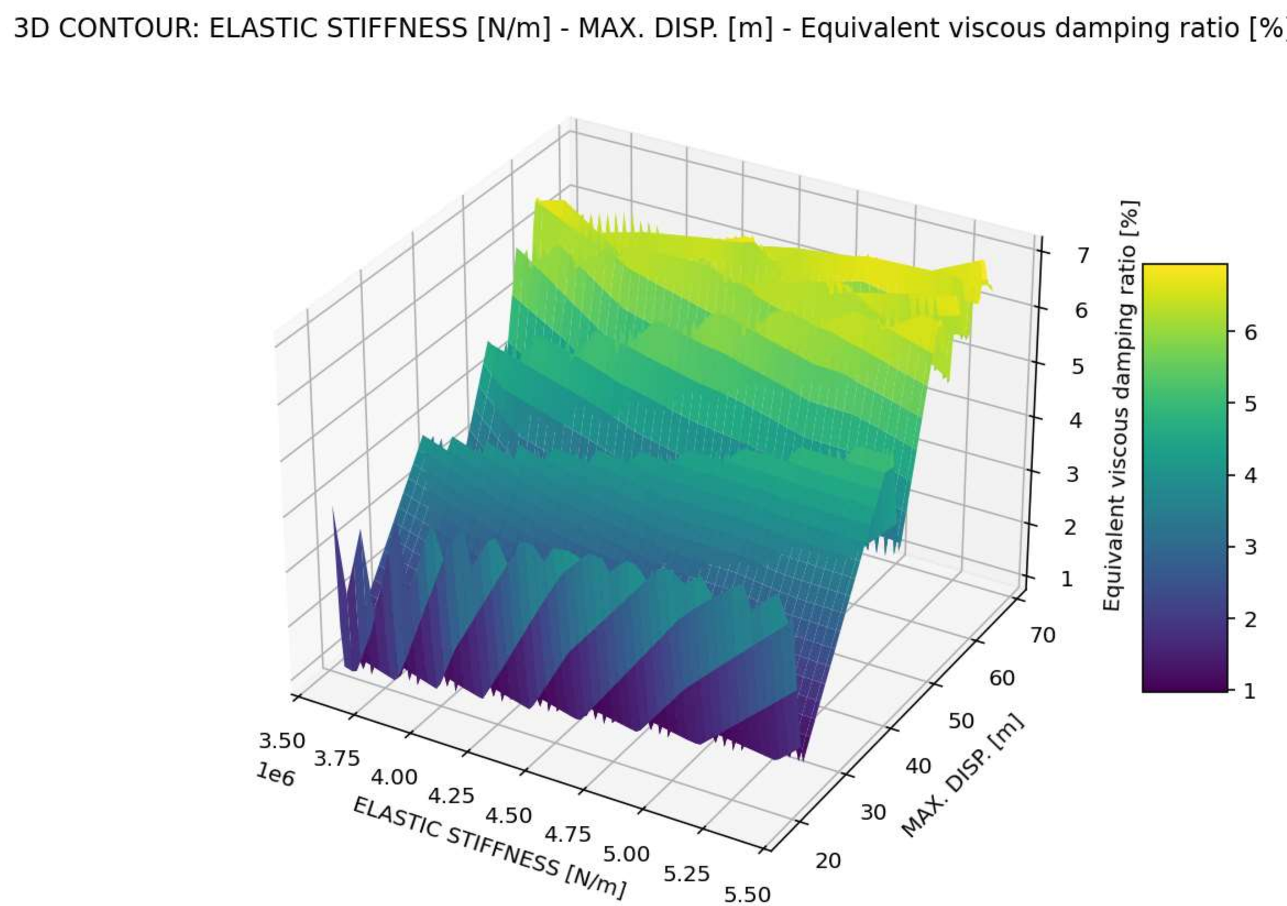



2D CONTOUR: ELASTIC STIFFNESS [N/m] – Topographic Map
MAX. ACCELERATION [m/s^2]
ELASTIC STIFFNESS [N/m]
1e6
Equivalent viscous damping ratio [%]
3.6
3.8
4.0
4.2
4.4
4.6
4.8
5.0
5.2
5.4
100000
150000
200000
250000
300000
0.6
1.5
2.4
3.3
4.2
5.1
6.0
6.9

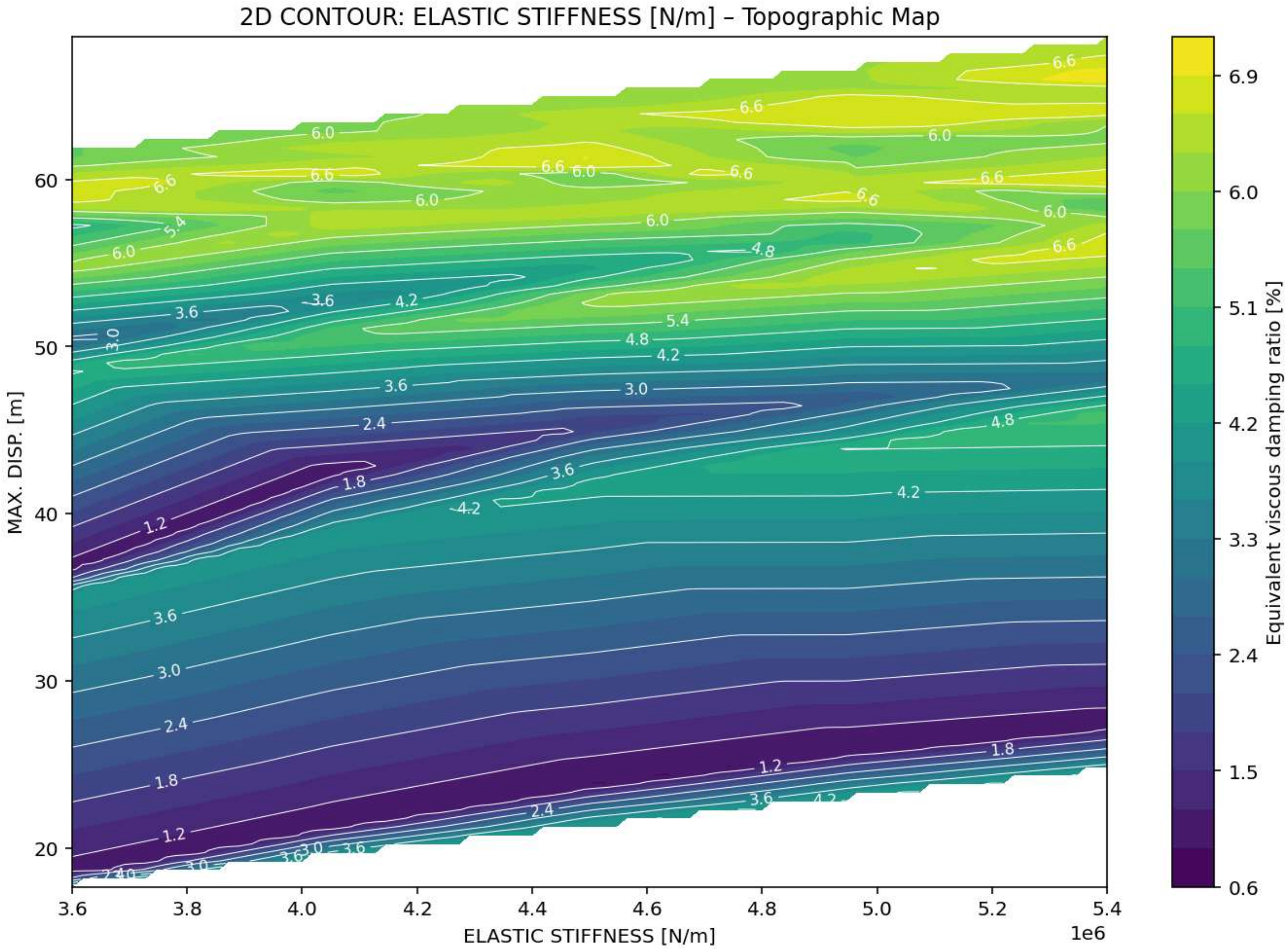


3D CONTOUR: ELASTIC STIFFNESS [N/m] - MAX. VELOCITY [m/s] - Equivalent viscous damping ratio [%]
ELASTIC STIFFNESS [N/m]
1e6
3.50
3.75
4.00
4.25
4.50
4.75
5.00
5.25
5.50
MAX. VELOCITY [m/s]
0.002
0.004
0.006
0.008
0.010
0.012
0.014
1
2
3
4
5
6
7
Equivalent viscous damping ratio [%]


2D CONTOUR: ELASTIC STIFFNESS [N/m] – Topographic Map
MAX. VELOCITY [m/s]
ELASTIC STIFFNESS [N/m]
1e6
3.6
3.8
4.0
4.2
4.4
4.6
4.8
5.0
5.2
5.4
0.004
0.006
0.008
0.010
0.012
1.8
2.4
3.0
3.6
4.2
4.8
5.4
6.0
6.6
Equivalent viscous damping ratio [%]
0.9
1.8
2.7
3.6
4.5
5.4
6.3
7.2

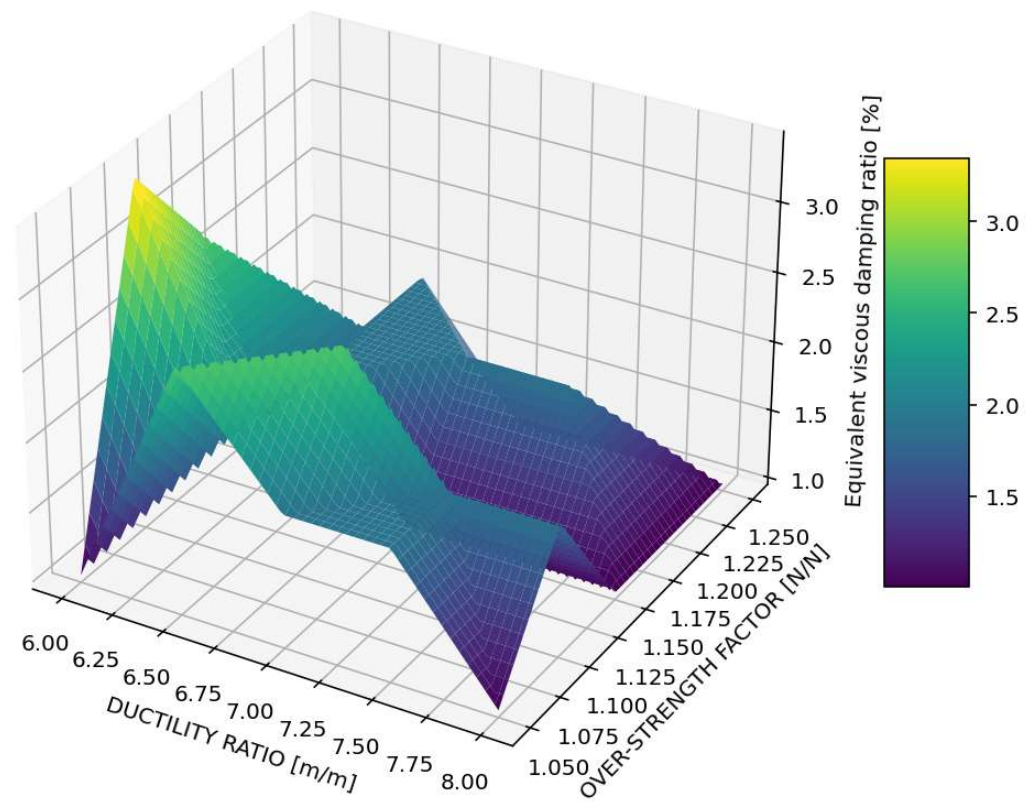


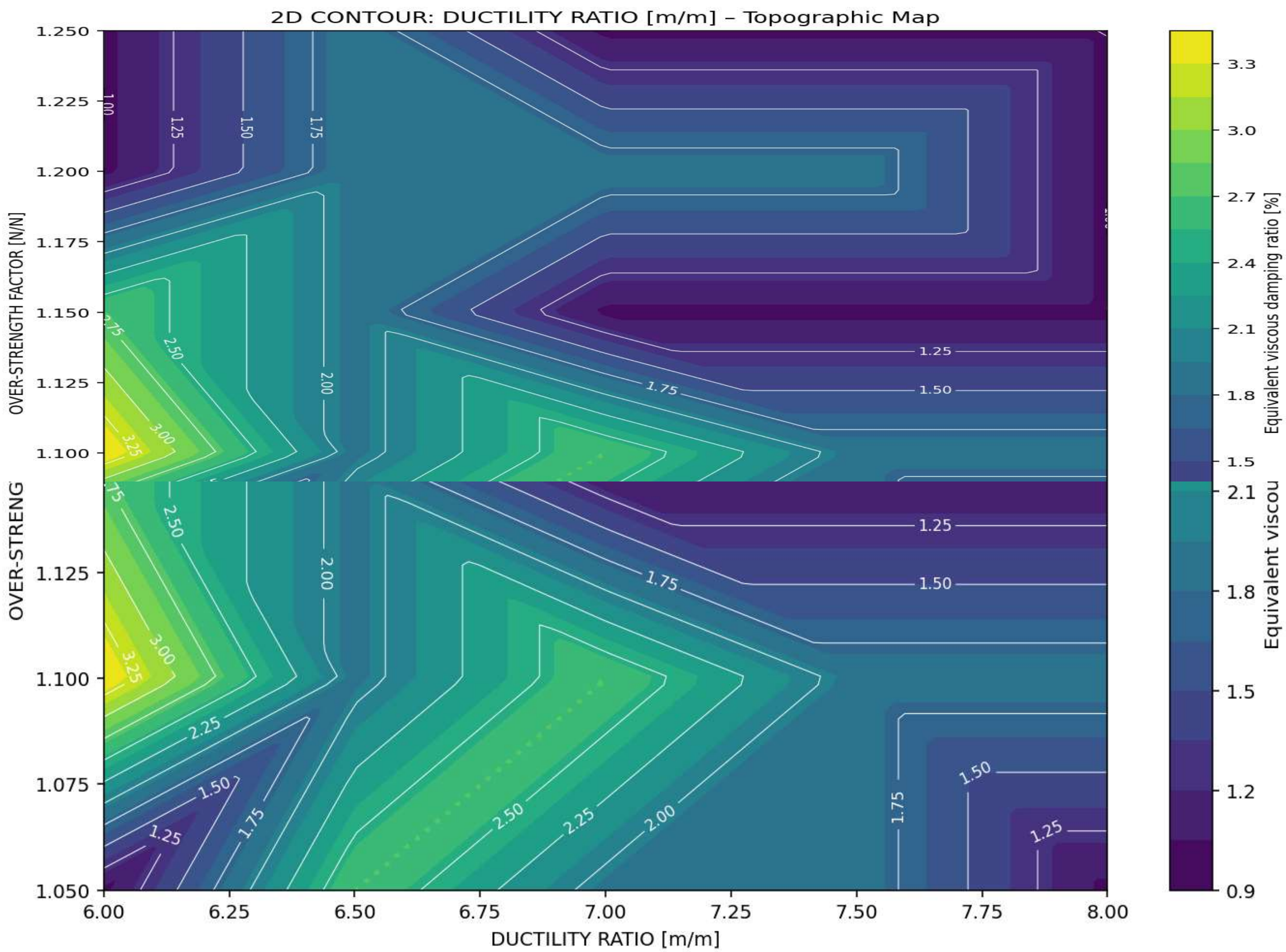
3D CONTOUR: ELASTIC STIFFNESS [N/m] - MAX. DISP. [m] - Equivalent viscous damping ratio [%]
ELASTIC STIFFNESS [N/m]
1e6
3.50
3.75
4.00
4.25
4.50
4.75
5.00
5.25
5.50
MAX. DISP. [m]
20
30
40
50
60
70
1
2
3
4
5
6
7
Equivalent viscous damping ratio [%]

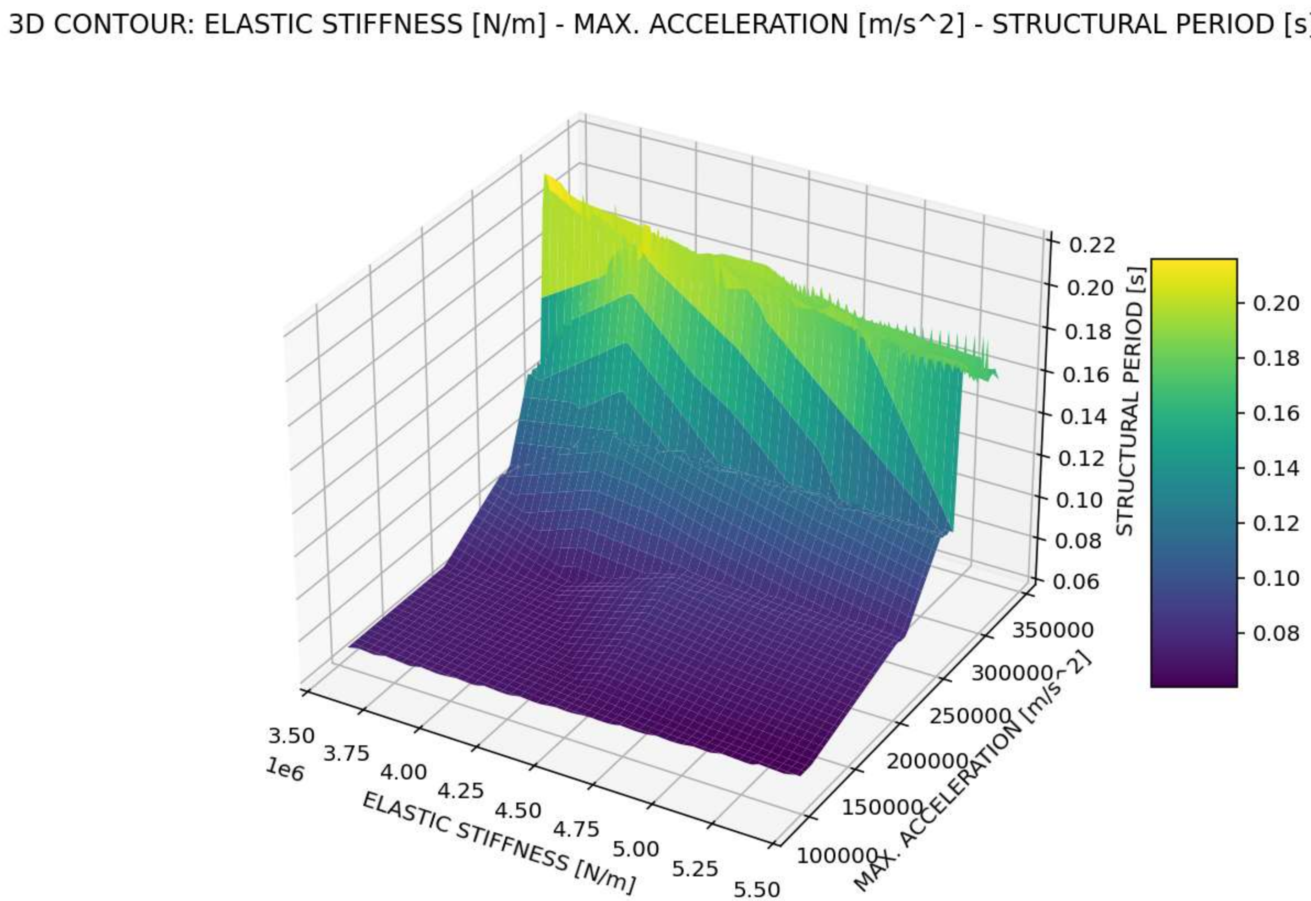


2D CONTOUR: ELASTIC STIFFNESS [N/m] – Topographic Map
MAX. DISP. [m]
ELASTIC STIFFNESS [N/m]
1e6
Equivalent viscous damping ratio [%]

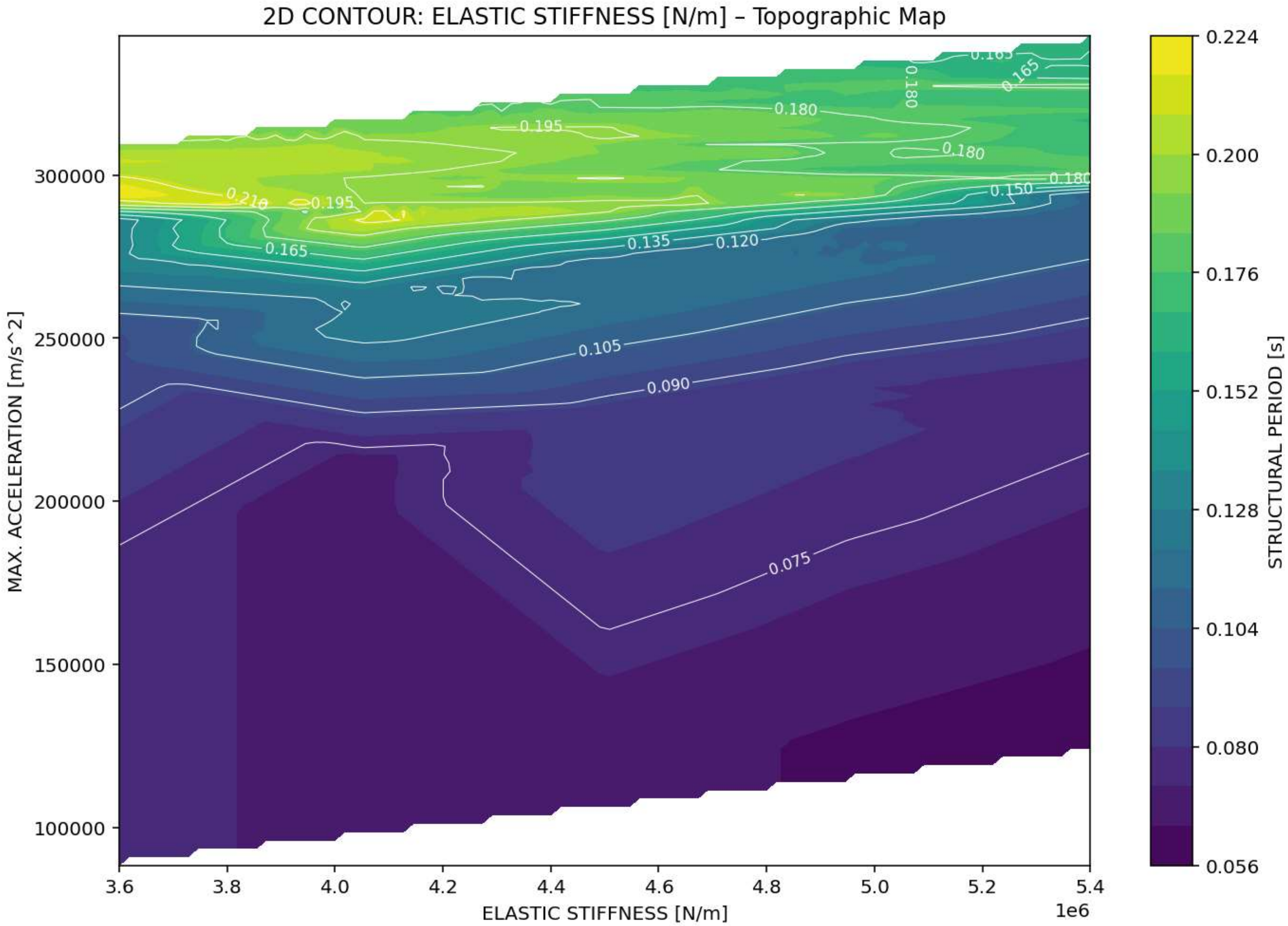
3D CONTOUR: DUCTILITY RATIO [m/m] - OVER-STRENGTH FACTOR [N/N] - Equivalent viscous damping ratio [%]



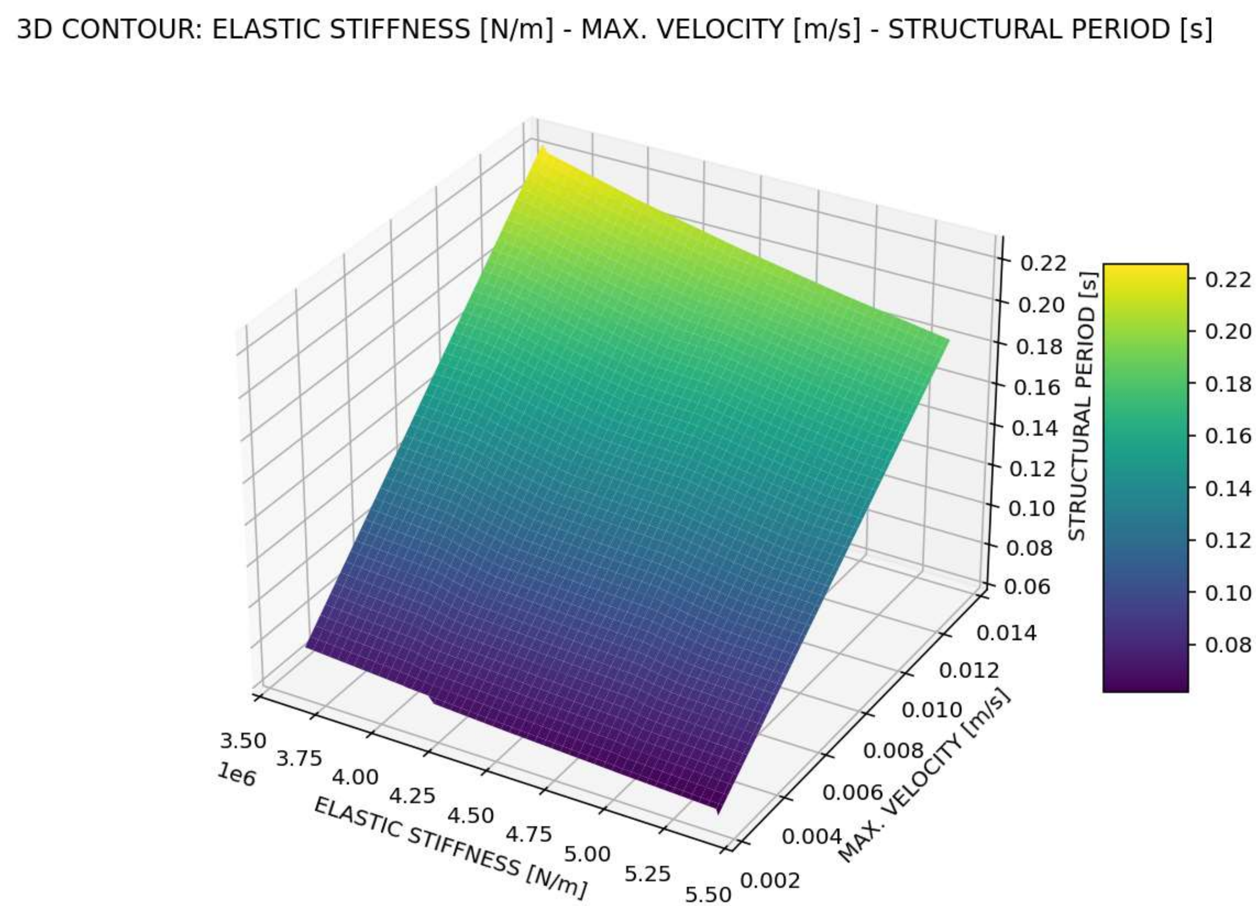

2D CONTOUR: DUCTILITY RATIO [m/m] – Topographic Map
OVER-STRENGTH FACTOR [N/N]
OVER-STRENG
DUCTILITY RATIO [m/m]
Equivalent viscous damping ratio [%]
Equivalent viscou

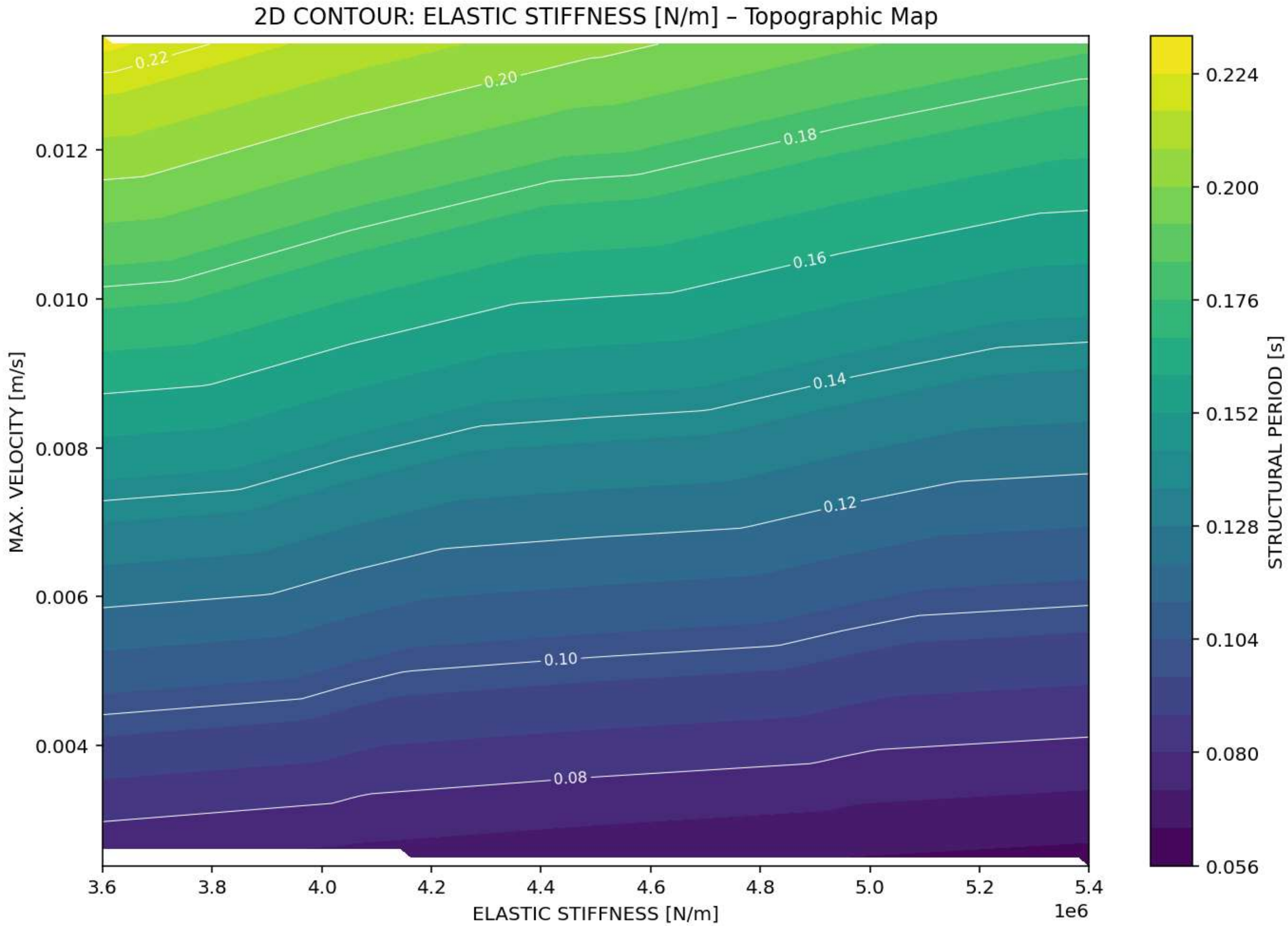


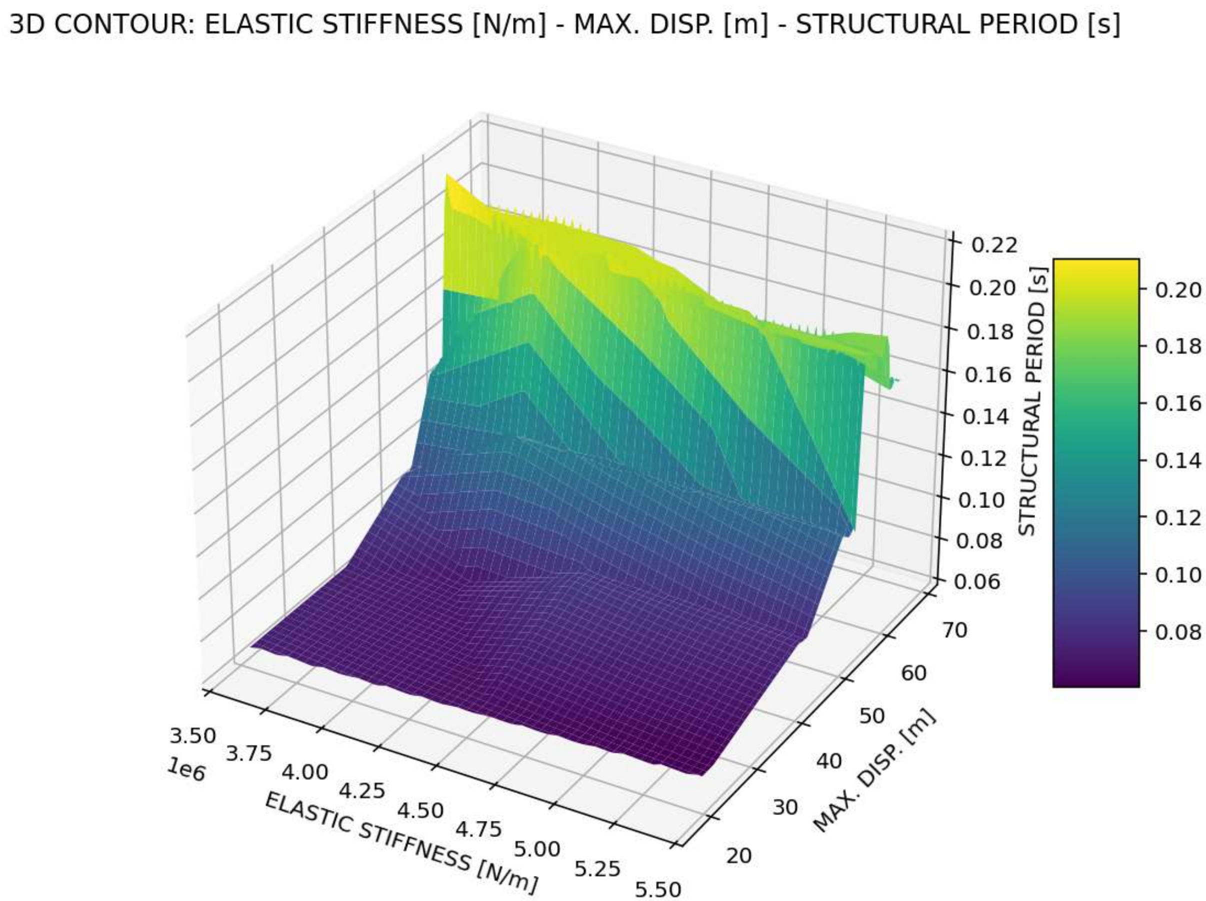
3D CONTOUR: ELASTIC STIFFNESS [N/m] - MAX. ACCELERATION [m/s^2] - STRUCTURAL PERIOD [s]
ELASTIC STIFFNESS [N/m]
1e6
3.50
3.75
4.00
4.25
4.50
4.75
5.00
5.25
5.50
MAX. ACCELERATION [m/s^2]
100000
150000
200000
250000
300000
350000
STRUCTURAL PERIOD [s]
0.06
0.08
0.10
0.12
0.14
0.16
0.18
0.20
0.22

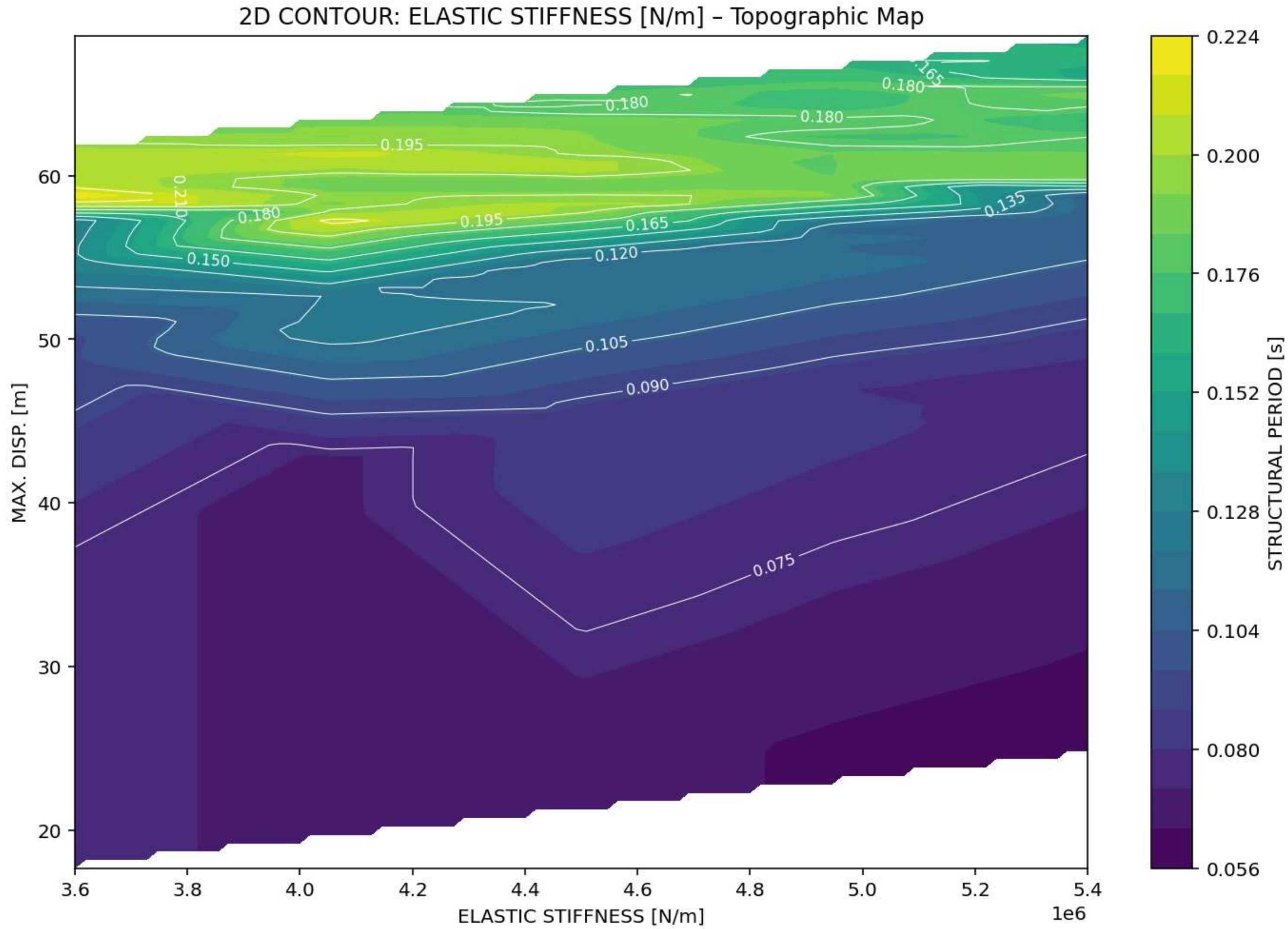


2D CONTOUR: ELASTIC STIFFNESS [N/m] – Topographic Map
MAX. ACCELERATION [m/s^2]
ELASTIC STIFFNESS [N/m]
1e6
STRUCTURAL PERIOD [s]
0.224
0.200
0.176
0.152
0.128
0.104
0.080
0.056
300000
250000
200000
150000
100000
3.6
3.8
4.0
4.2
4.4
4.6
4.8
5.0
5.2
5.4
0.210
0.195
0.180
0.165
0.150
0.135
0.120
0.105
0.090
0.075

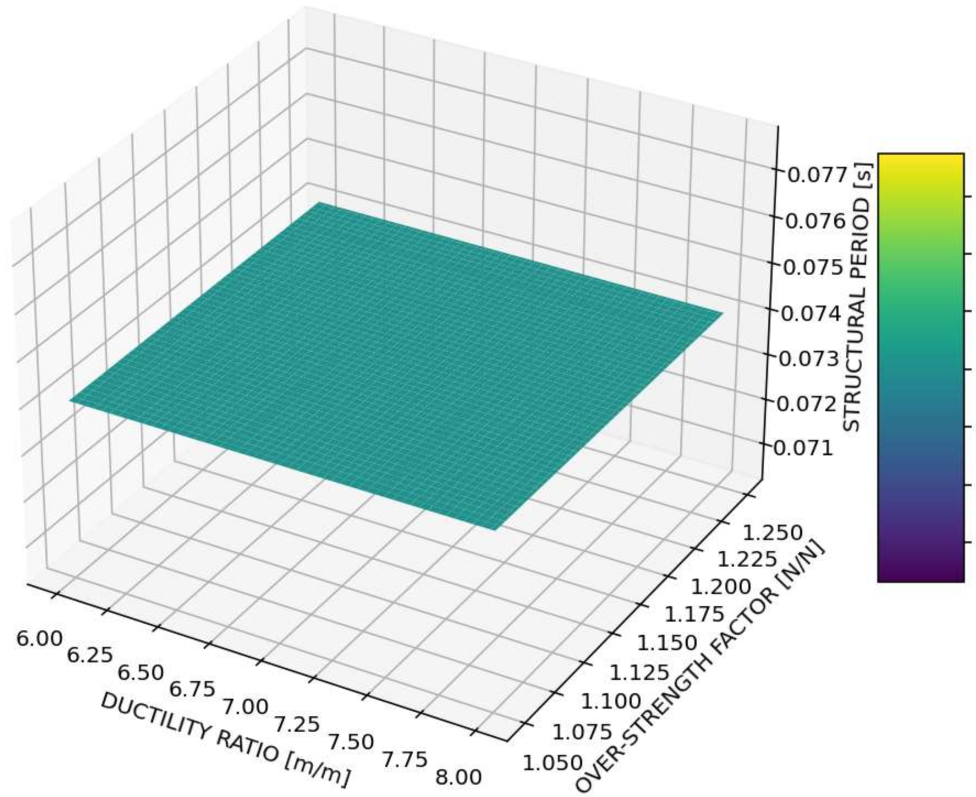

3D CONTOUR: ELASTIC STIFFNESS [N/m] - MAX. VELOCITY [m/s] - STRUCTURAL PERIOD [s]
ELASTIC STIFFNESS [N/m]
1e6
3.50
3.75
4.00
4.25
4.50
4.75
5.00
5.25
5.50
MAX. VELOCITY [m/s]
0.002
0.004
0.006
0.008
0.010
0.012
0.014
STRUCTURAL PERIOD [s]
0.06
0.08
0.10
0.12
0.14
0.16
0.18
0.20
0.22

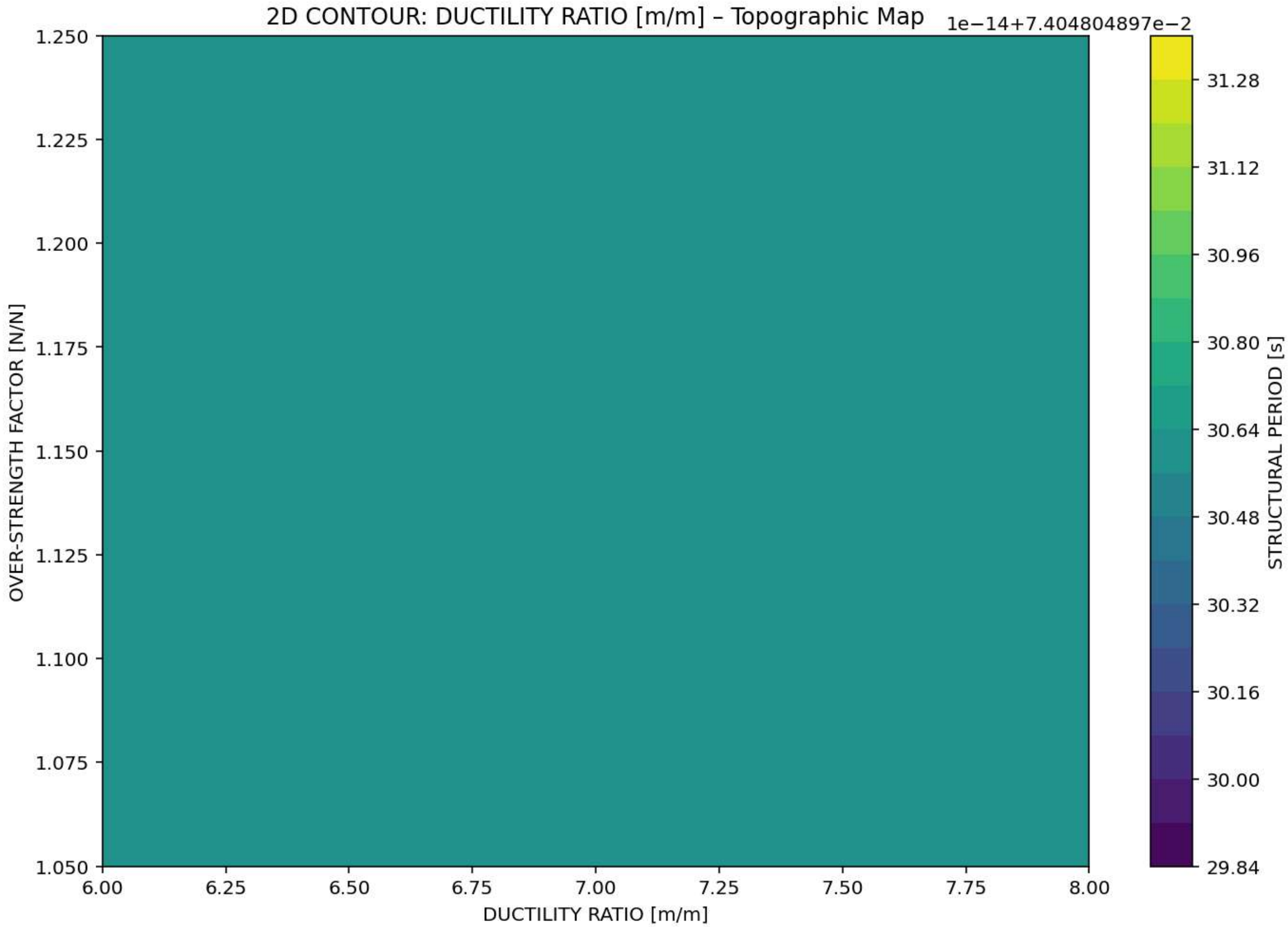

2D CONTOUR: ELASTIC STIFFNESS [N/m] – Topographic Map
0.22
0.20
0.18
0.16
0.14
0.12
0.10
0.08
MAX. VELOCITY [m/s]
0.012
0.010
0.008
0.006
0.004
ELASTIC STIFFNESS [N/m]
3.6
3.8
4.0
4.2
4.4
4.6
4.8
5.0
5.2
5.4
1e6
STRUCTURAL PERIOD [s]
0.224
0.200
0.176
0.152
0.128
0.104
0.080
0.056

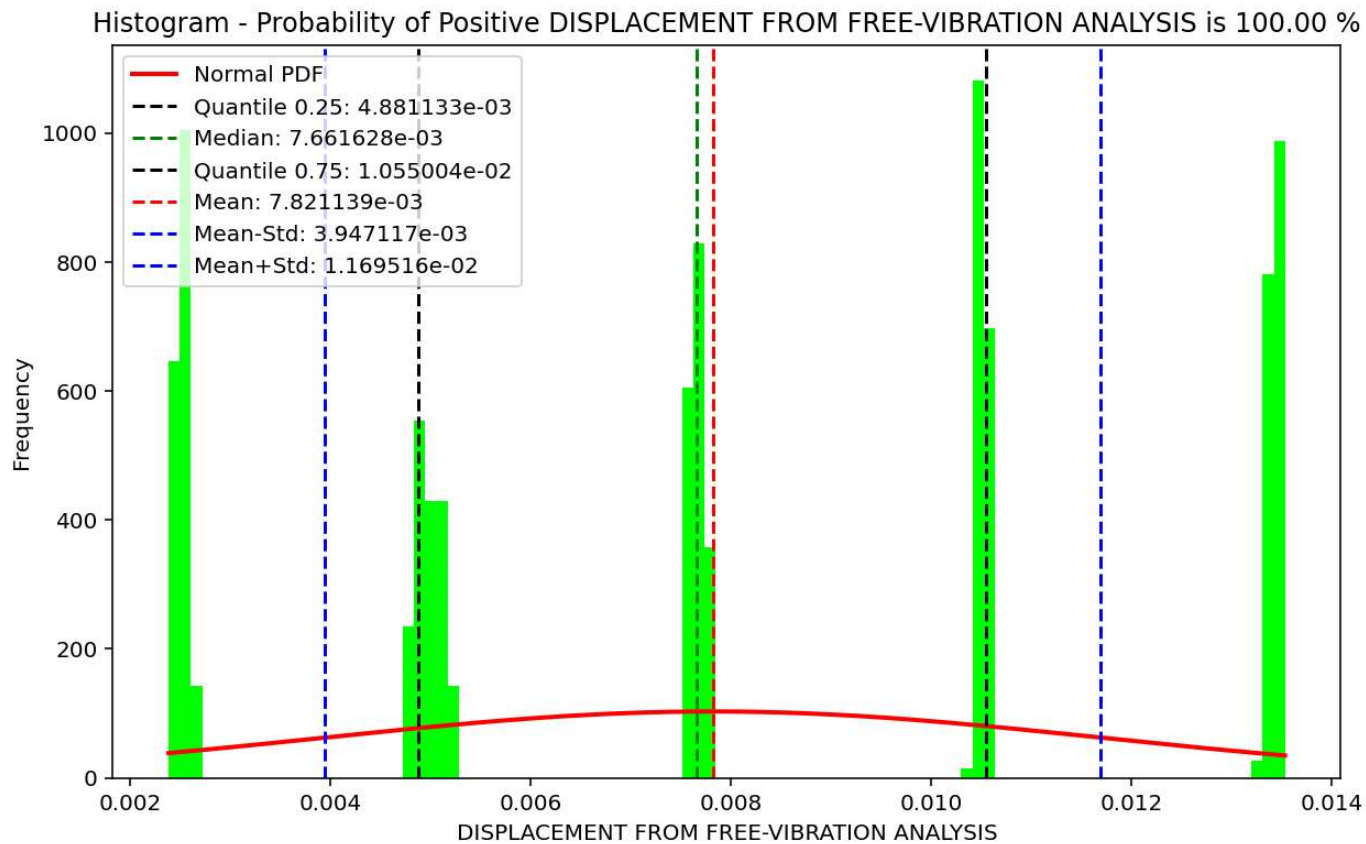

3D CONTOUR: ELASTIC STIFFNESS [N/m] - MAX. DISP. [m] - STRUCTURAL PERIOD [s]
ELASTIC STIFFNESS [N/m]
1e6
3.50
3.75
4.00
4.25
4.50
4.75
5.00
5.25
5.50
MAX. DISP. [m]
20
30
40
50
60
70
STRUCTURAL PERIOD [s]
0.06
0.08
0.10
0.12
0.14
0.16
0.18
0.20
0.22

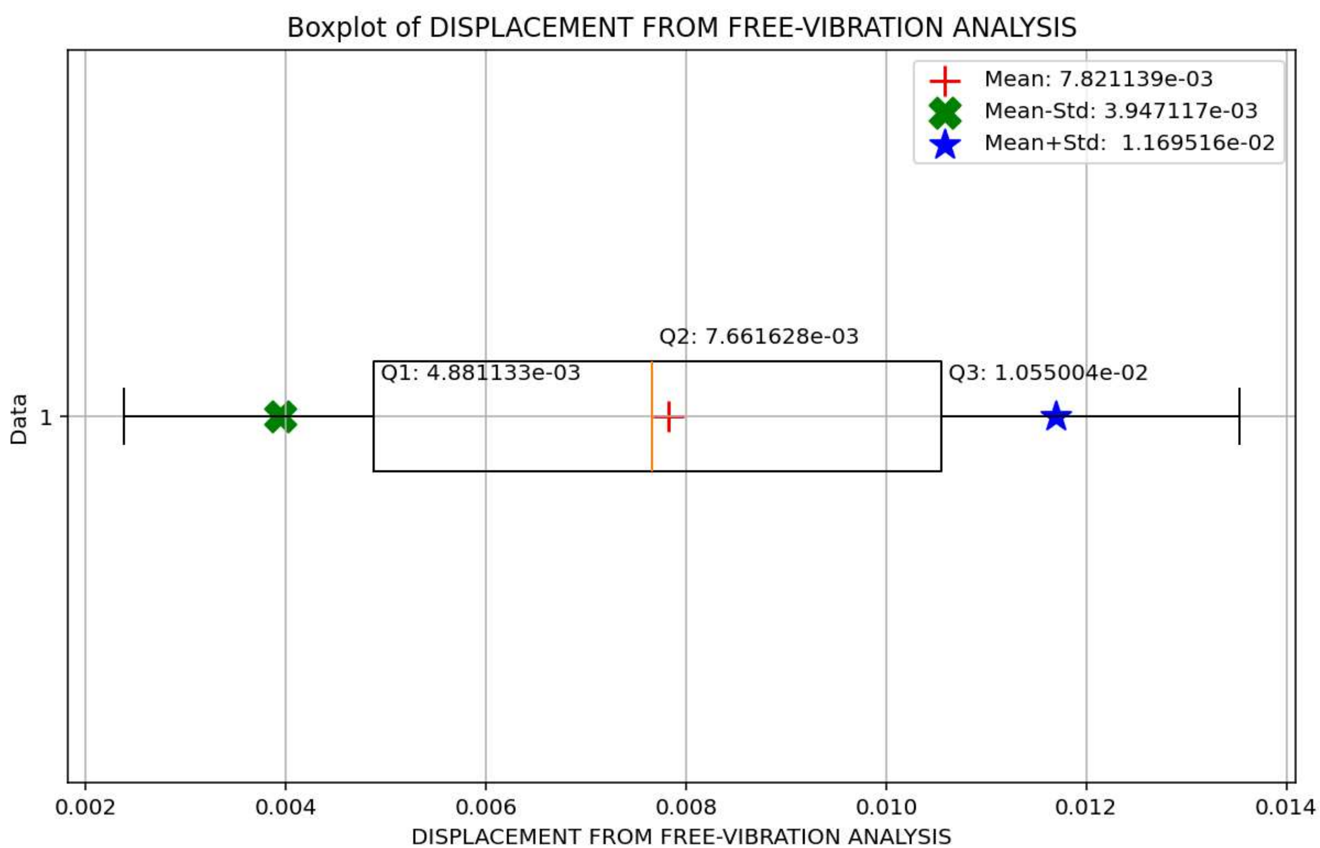

2D CONTOUR: ELASTIC STIFFNESS [N/m] – Topographic Map
MAX. DISP. [m]
ELASTIC STIFFNESS [N/m]
1e6
STRUCTURAL PERIOD [s]
0.224
0.200
0.176
0.152
0.128
0.104
0.080
0.056
0.210
0.195
0.180
0.165
0.150
0.135
0.120
0.105
0.090
0.075
3.6
3.8
4.0
4.2
4.4
4.6
4.8
5.0
5.2
5.4
20
30
40
50
60

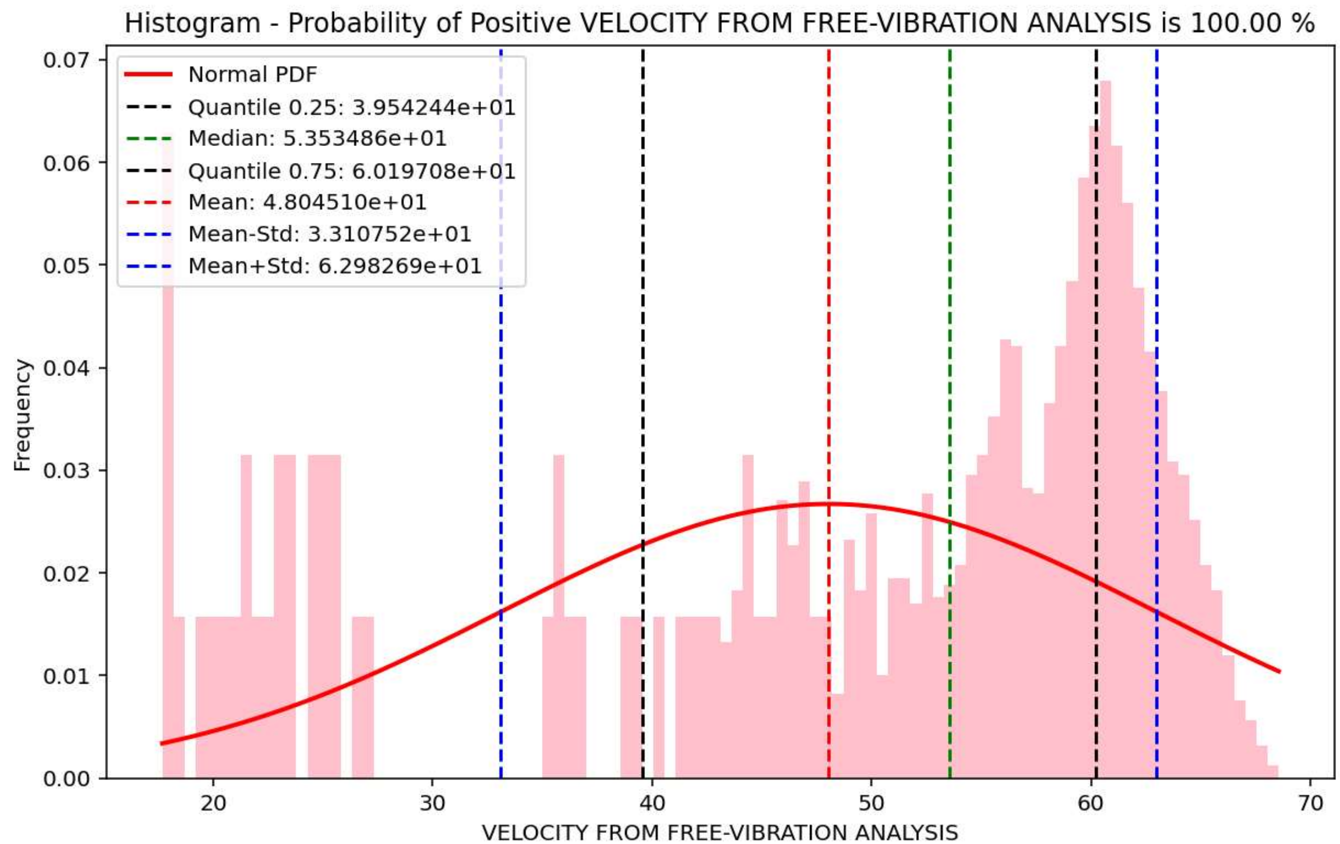
3D CONTOUR: DUCTILITY RATIO [m/m] - OVER-STRENGTH FACTOR [N/N] - STRUCTURAL PERIOD [s]



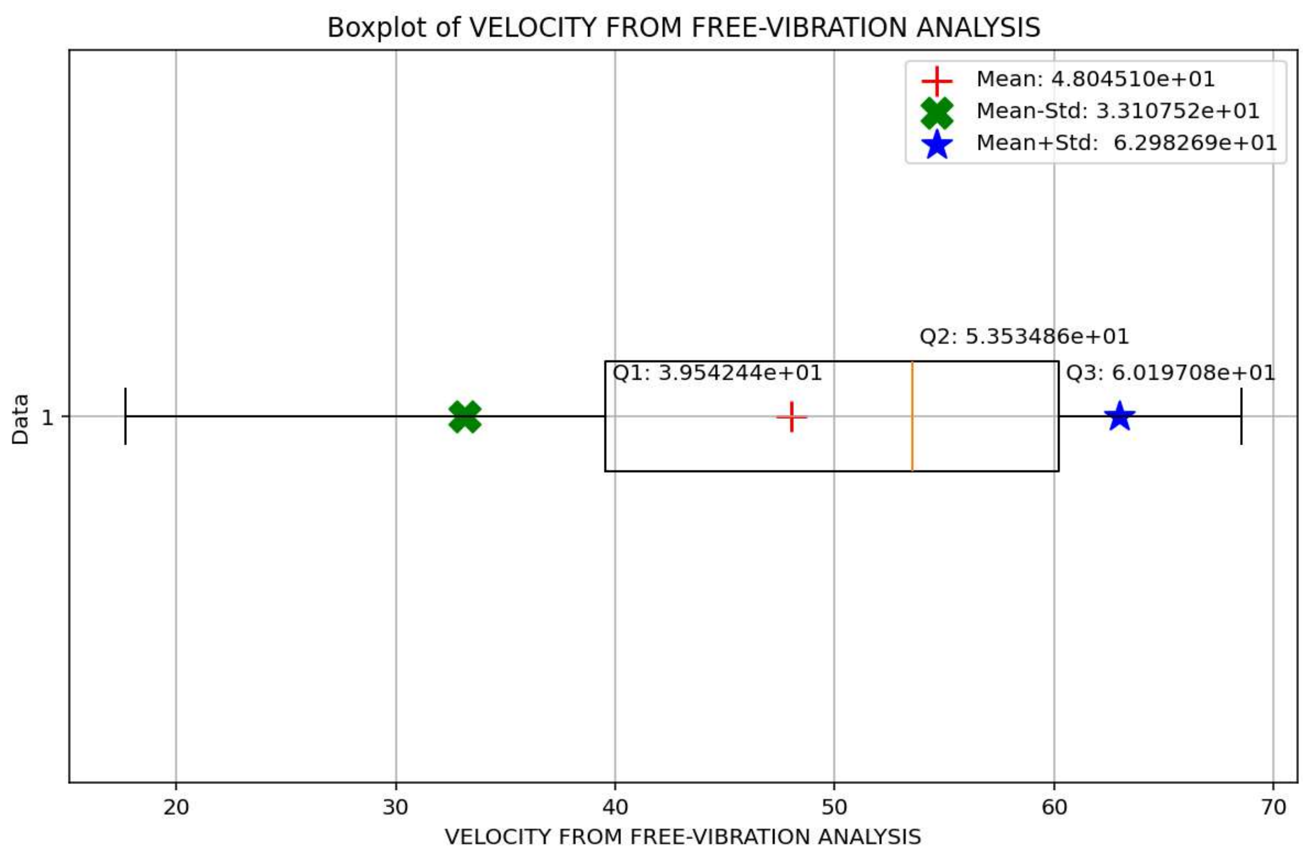

2D CONTOUR: DUCTILITY RATIO [m/m] – Topographic Map
1e−14+7.404804897e−2
OVER-STRENGTH FACTOR [N/N]
1.250
1.225
1.200
1.175
1.150
1.125
1.100
1.075
1.050
6.00
6.25
6.50
6.75
7.00
7.25
7.50
7.75
8.00
DUCTILITY RATIO [m/m]
STRUCTURAL PERIOD [s]
31.28
31.12
30.96
30.80
30.64
30.48
30.32
30.16
30.00
29.84

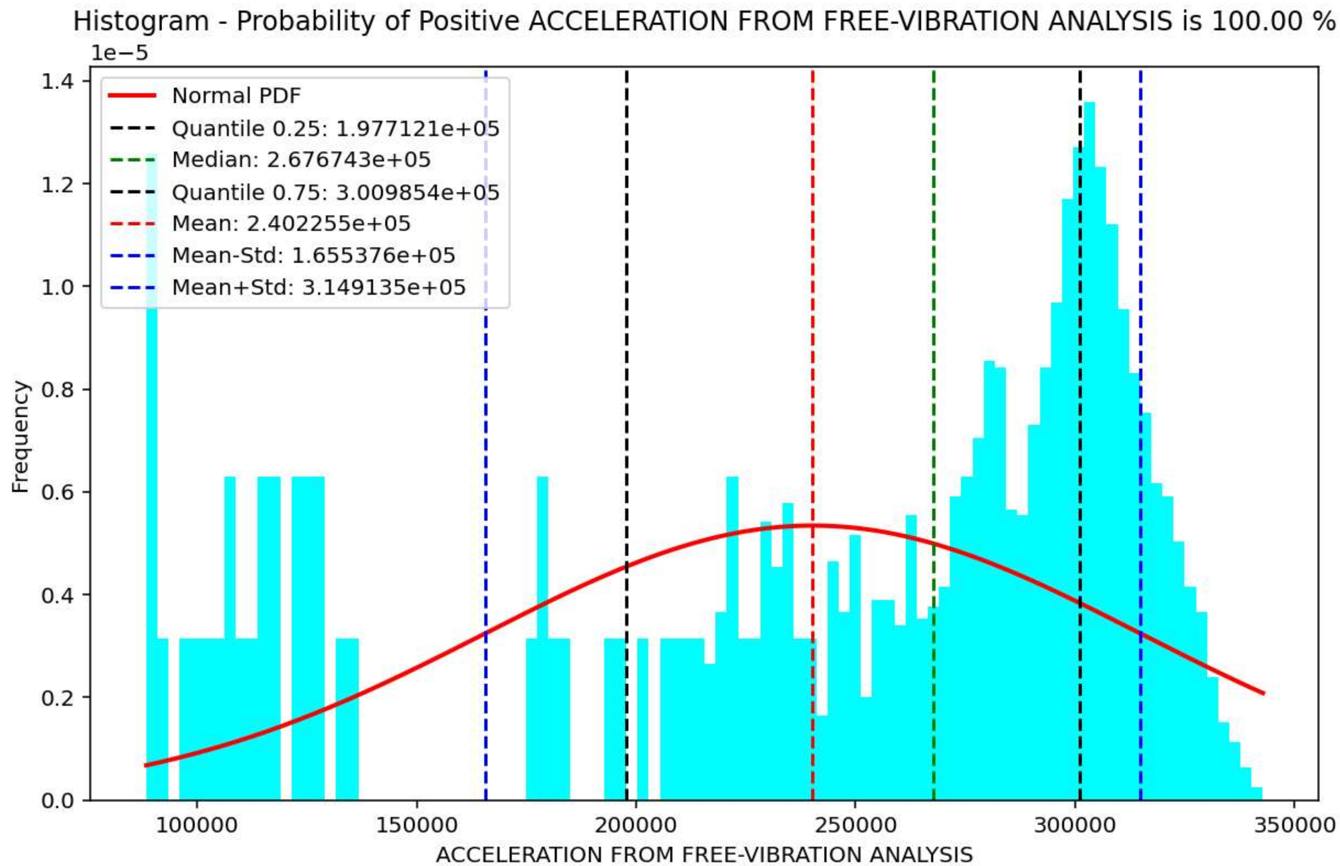

Histogram - Probability of Positive DISPLACEMENT FROM FREE-VIBRATION ANALYSIS is 100.00 %
Normal PDF
Quantile 0.25: 4.881133e-03
Median: 7.661628e-03
Quantile 0.75: 1.055004e-02
Mean: 7.821139e-03
Mean-Std: 3.947117e-03
Mean+Std: 1.169516e-02
Frequency
0
200
400
600
800
1000
0.002
0.004
0.006
0.008
0.010
0.012
0.014
DISPLACEMENT FROM FREE-VIBRATION ANALYSIS

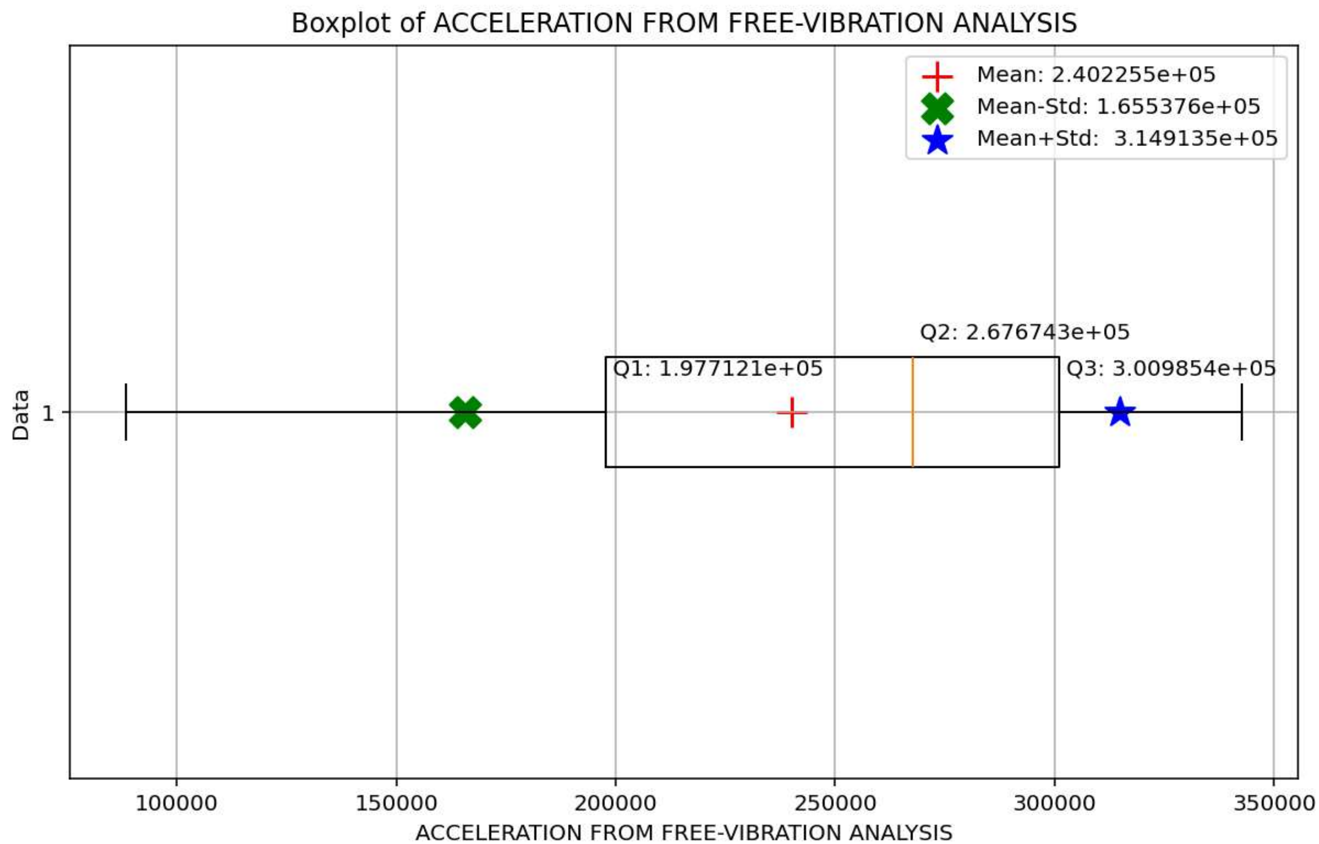

Boxplot of DISPLACEMENT FROM FREE-VIBRATION ANALYSIS
Mean: 7.821139e-03
Mean-Std: 3.947117e-03
Mean+Std: 1.169516e-02
Q1: 4.881133e-03
Q2: 7.661628e-03
Q3: 1.055004e-02
Data
1
0.002
0.004
0.006
0.008
0.010
0.012
0.014
DISPLACEMENT FROM FREE-VIBRATION ANALYSIS

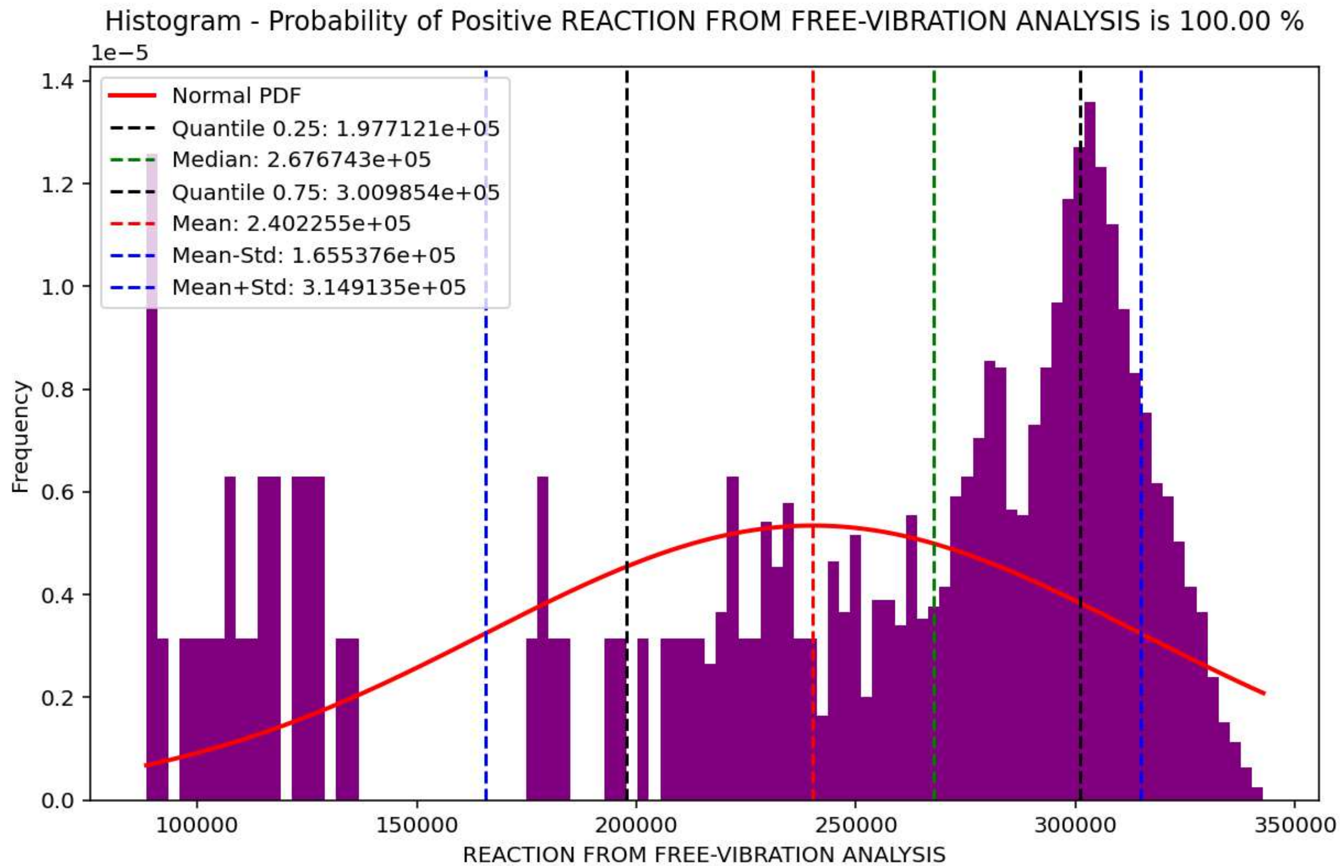


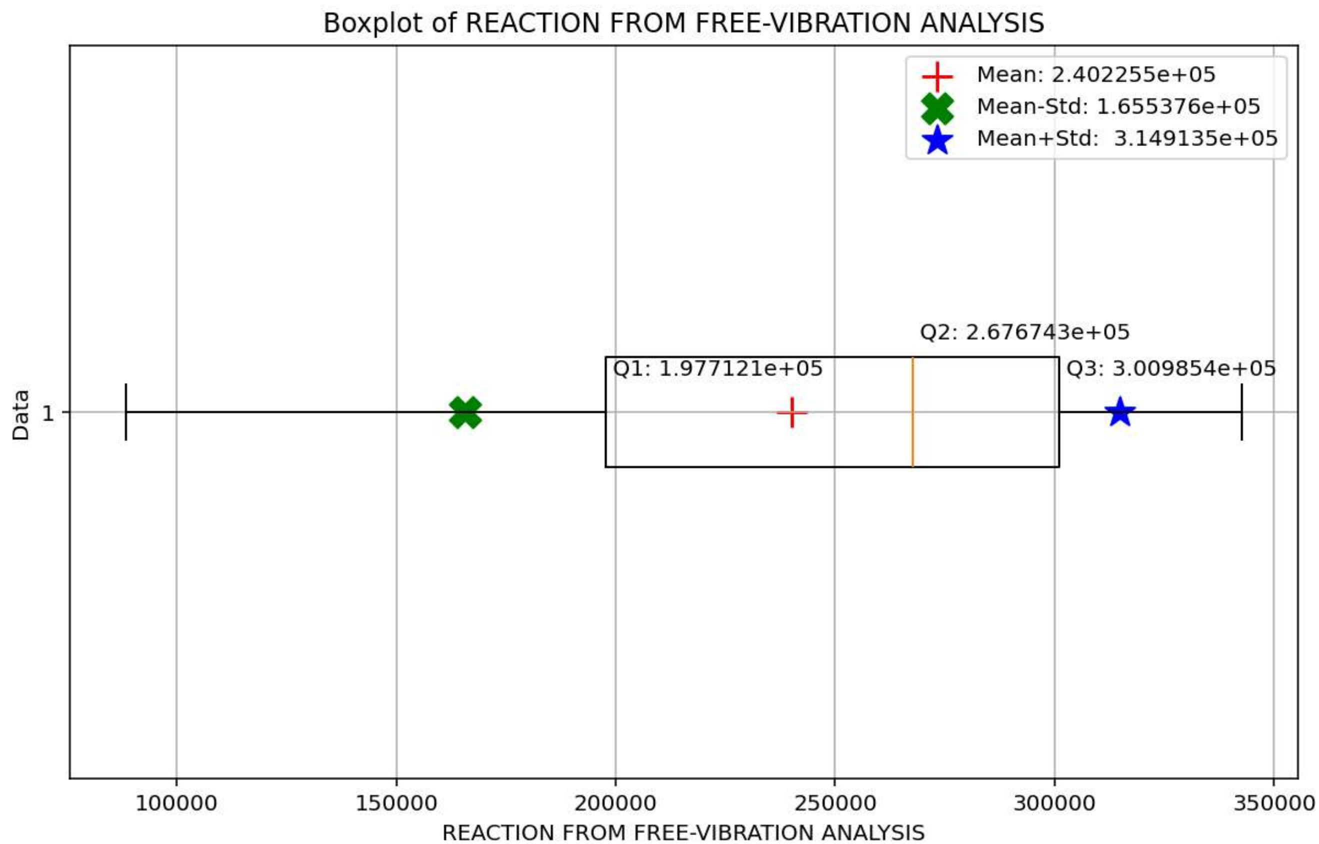
Histogram - Probability of Positive VELOCITY FROM FREE-VIBRATION ANALYSIS is 100.00 %
Normal PDF
Quantile 0.25: 3.954244e+01
Median: 5.353486e+01
Quantile 0.75: 6.019708e+01
Mean: 4.804510e+01
Mean-Std: 3.310752e+01
Mean+Std: 6.298269e+01
Frequency
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
20
30
40
50
60
70
VELOCITY FROM FREE-VIBRATION ANALYSIS

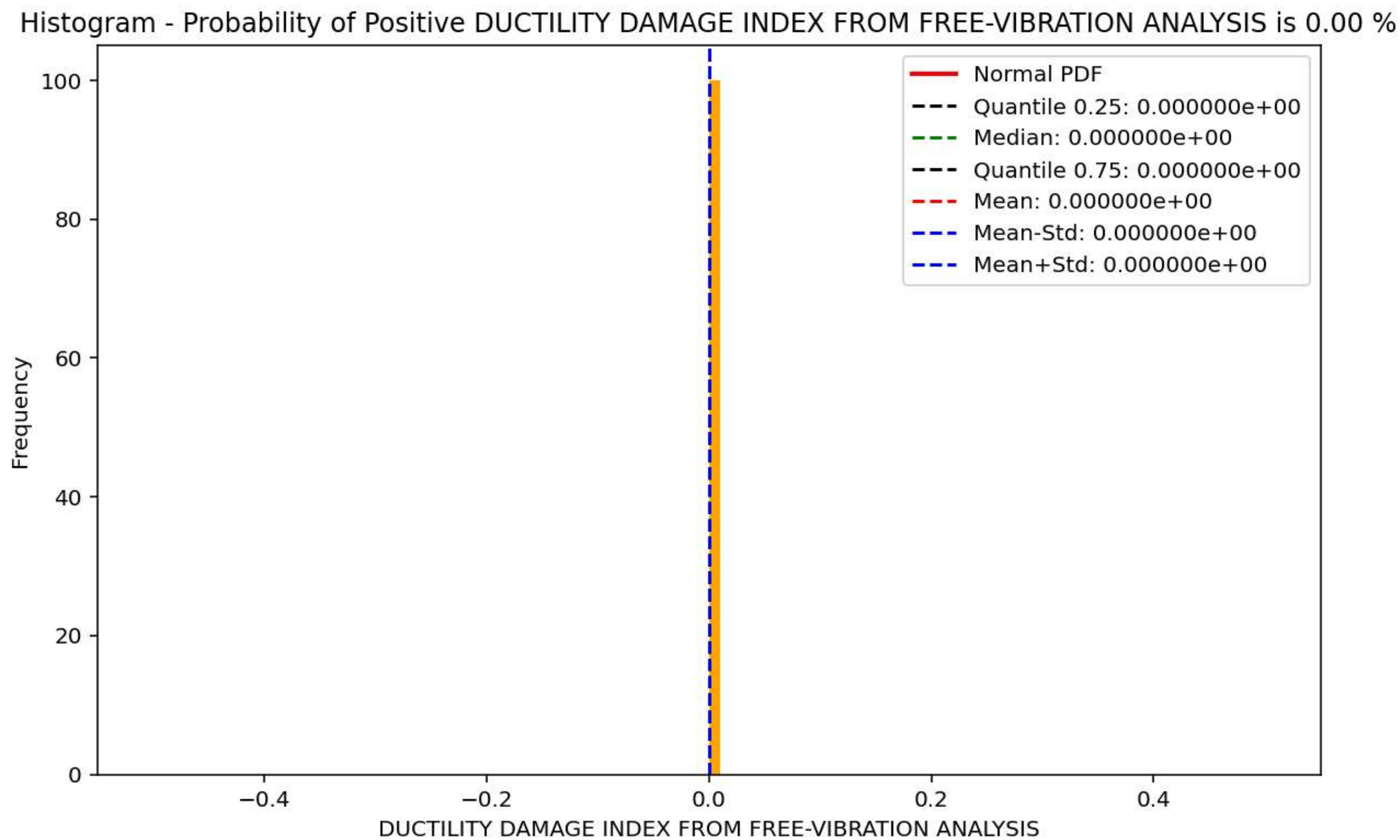

Boxplot of VELOCITY FROM FREE-VIBRATION ANALYSIS
Mean: 4.804510e+01
Mean-Std: 3.310752e+01
Mean+Std: 6.298269e+01
Q1: 3.954244e+01
Q2: 5.353486e+01
Q3: 6.019708e+01
Data
1
20
30
40
50
60
70
VELOCITY FROM FREE-VIBRATION ANALYSIS


Histogram - Probability of Positive ACCELERATION FROM FREE-VIBRATION ANALYSIS is 100.00 %
1e−5
Normal PDF
Quantile 0.25: 1.977121e+05
Median: 2.676743e+05
Quantile 0.75: 3.009854e+05
Mean: 2.402255e+05
Mean-Std: 1.655376e+05
Mean+Std: 3.149135e+05
Frequency
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
100000
150000
200000
250000
300000
350000
ACCELERATION FROM FREE-VIBRATION ANALYSIS

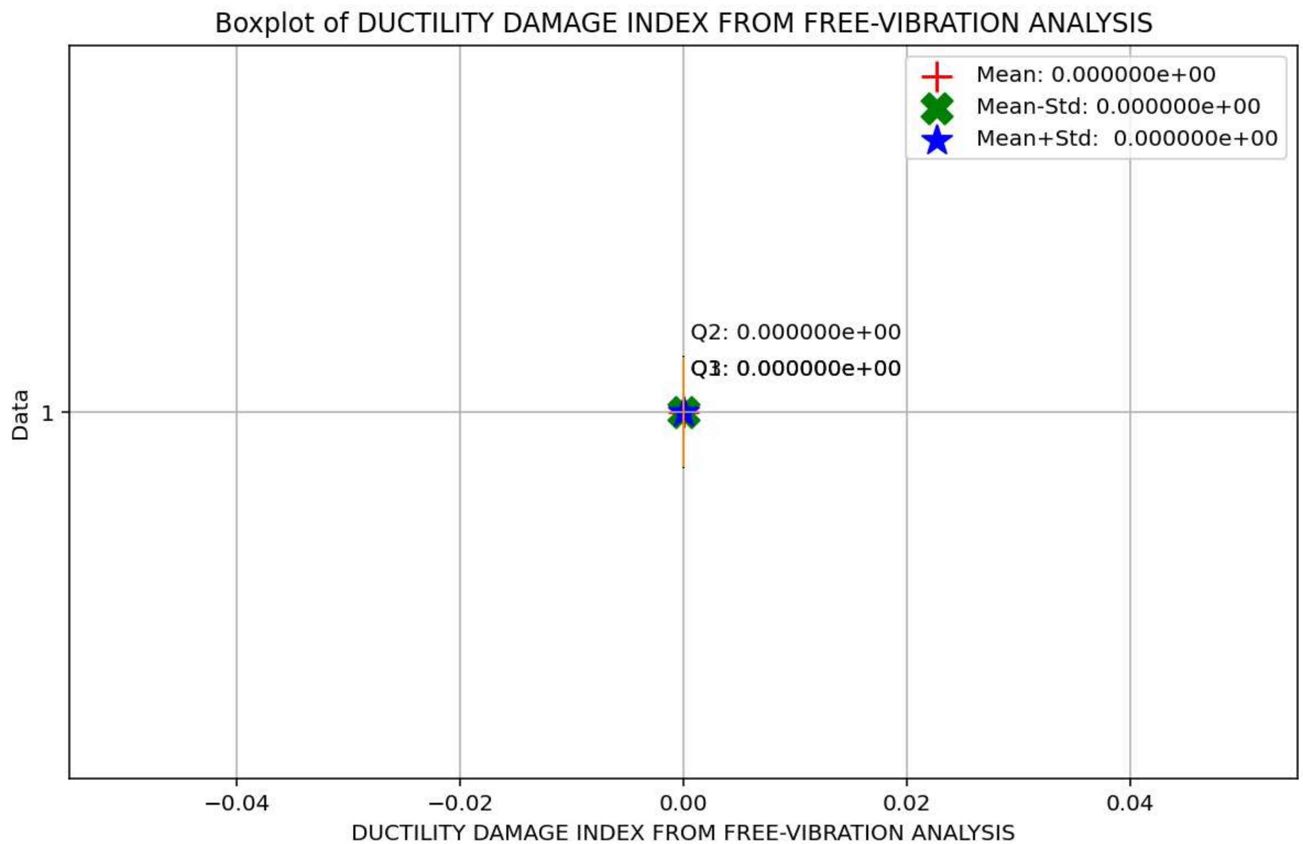

Boxplot of ACCELERATION FROM FREE-VIBRATION ANALYSIS
Mean: 2.402255e+05
Mean-Std: 1.655376e+05
Mean+Std: 3.149135e+05
Q2: 2.676743e+05
Q1: 1.977121e+05
Q3: 3.009854e+05
Data
1
100000
150000
200000
250000
300000
350000
ACCELERATION FROM FREE-VIBRATION ANALYSIS

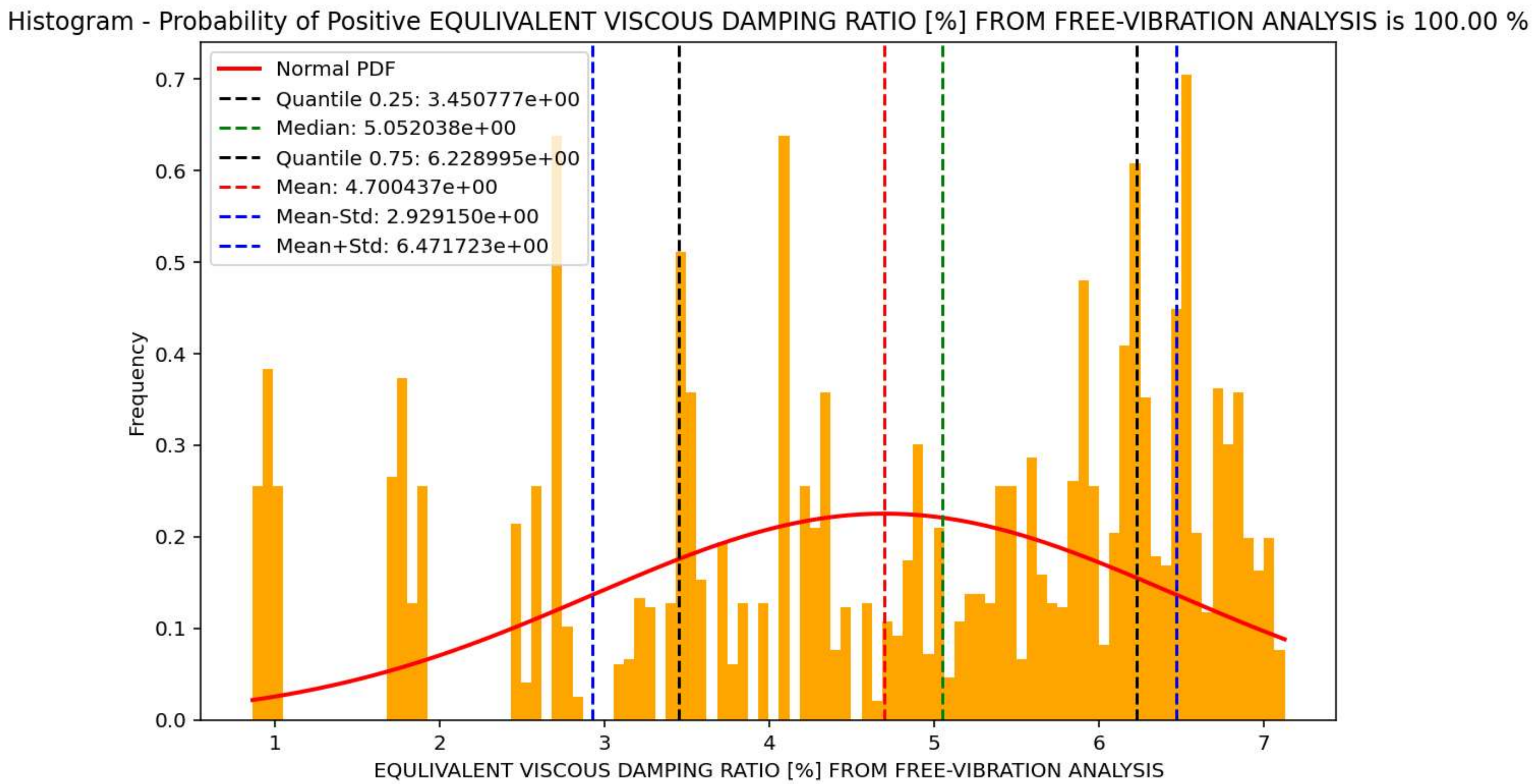

Histogram - Probability of Positive REACTION FROM FREE-VIBRATION ANALYSIS is 100.00 %
1e−5
Normal PDF
Quantile 0.25: 1.977121e+05
Median: 2.676743e+05
Quantile 0.75: 3.009854e+05
Mean: 2.402255e+05
Mean-Std: 1.655376e+05
Mean+Std: 3.149135e+05
Frequency
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
100000
150000
200000
250000
300000
350000
REACTION FROM FREE-VIBRATION ANALYSIS

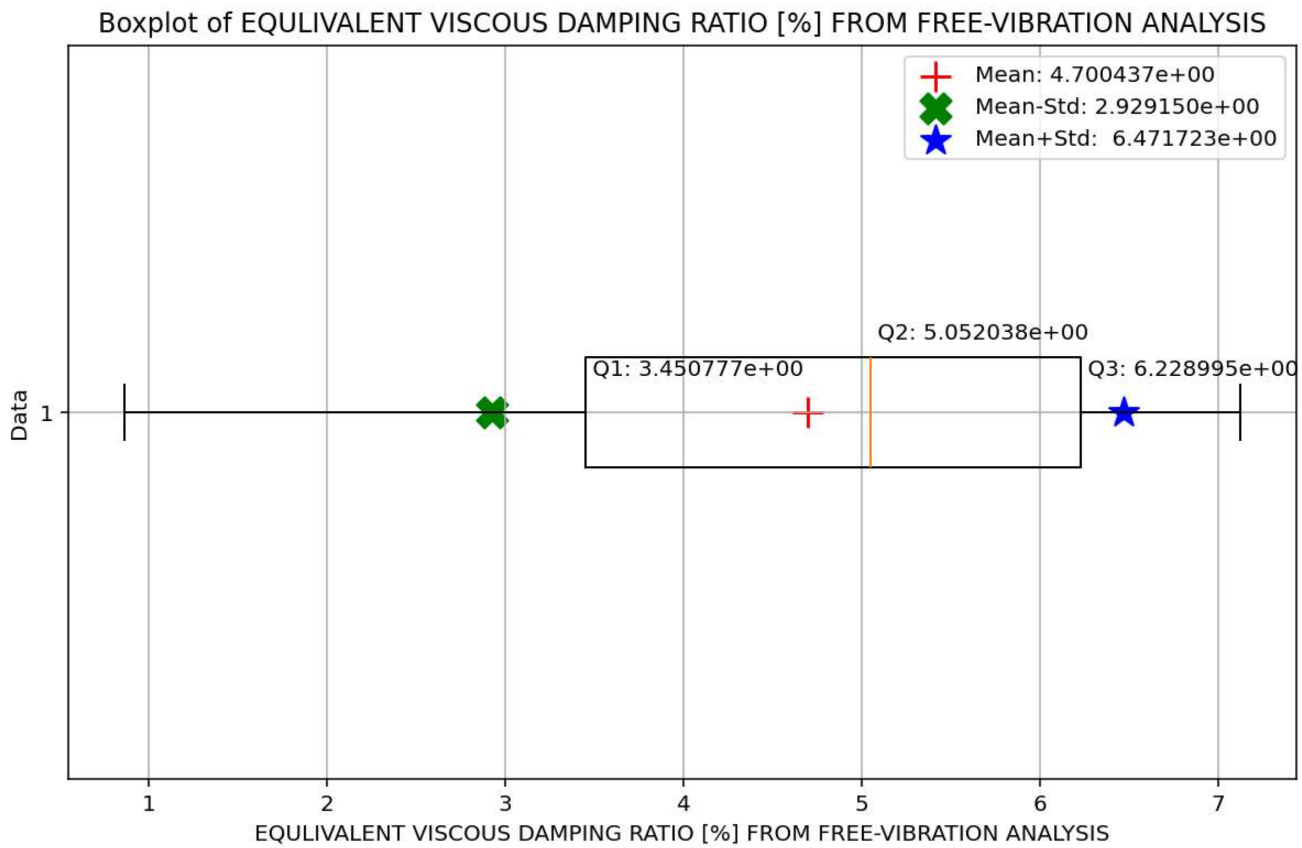

Boxplot of REACTION FROM FREE-VIBRATION ANALYSIS
Mean: 2.402255e+05
Mean-Std: 1.655376e+05
Mean+Std: 3.149135e+05
Q1: 1.977121e+05
Q2: 2.676743e+05
Q3: 3.009854e+05
Data
1
100000
150000
200000
250000
300000
350000
REACTION FROM FREE-VIBRATION ANALYSIS

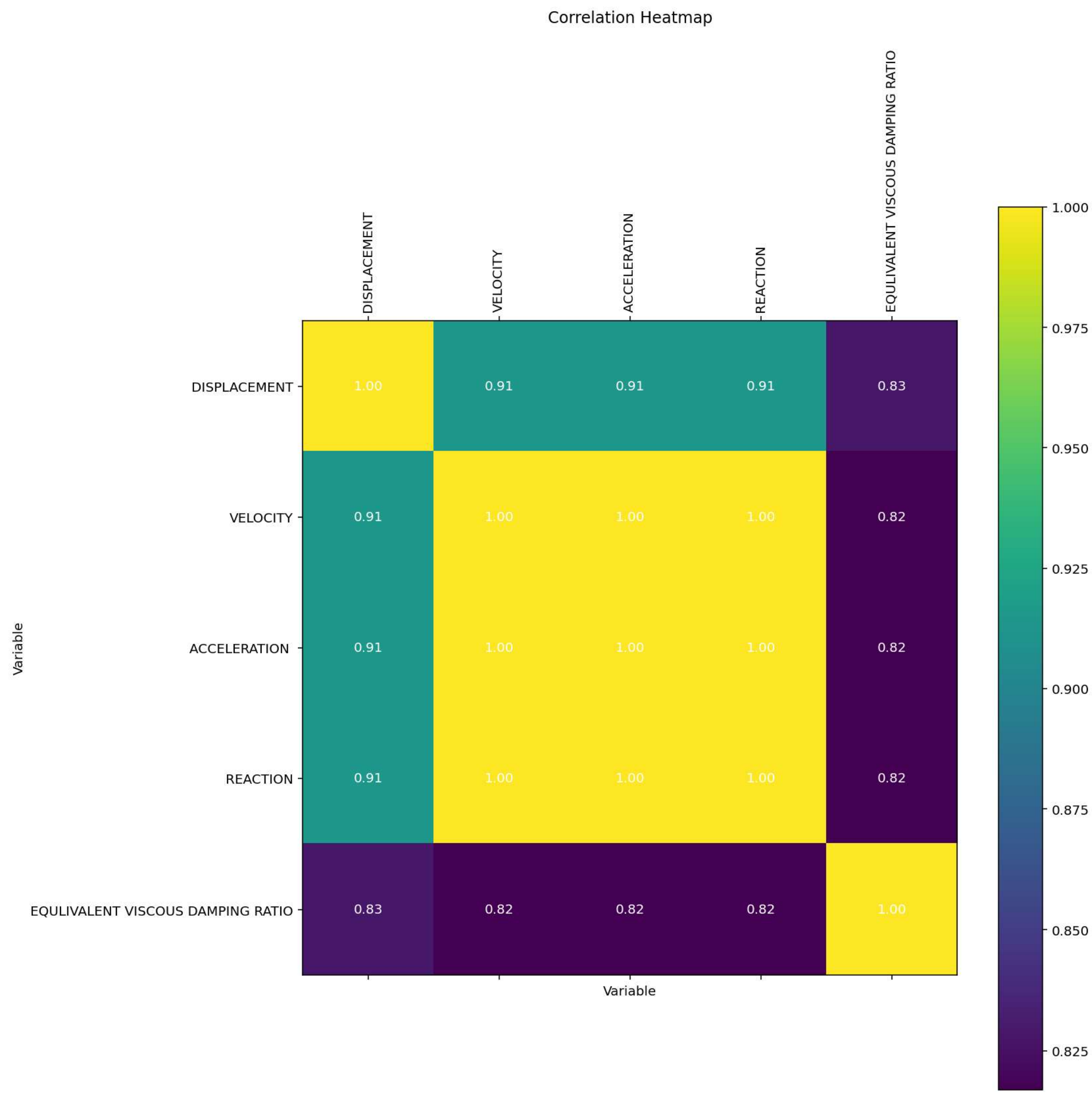


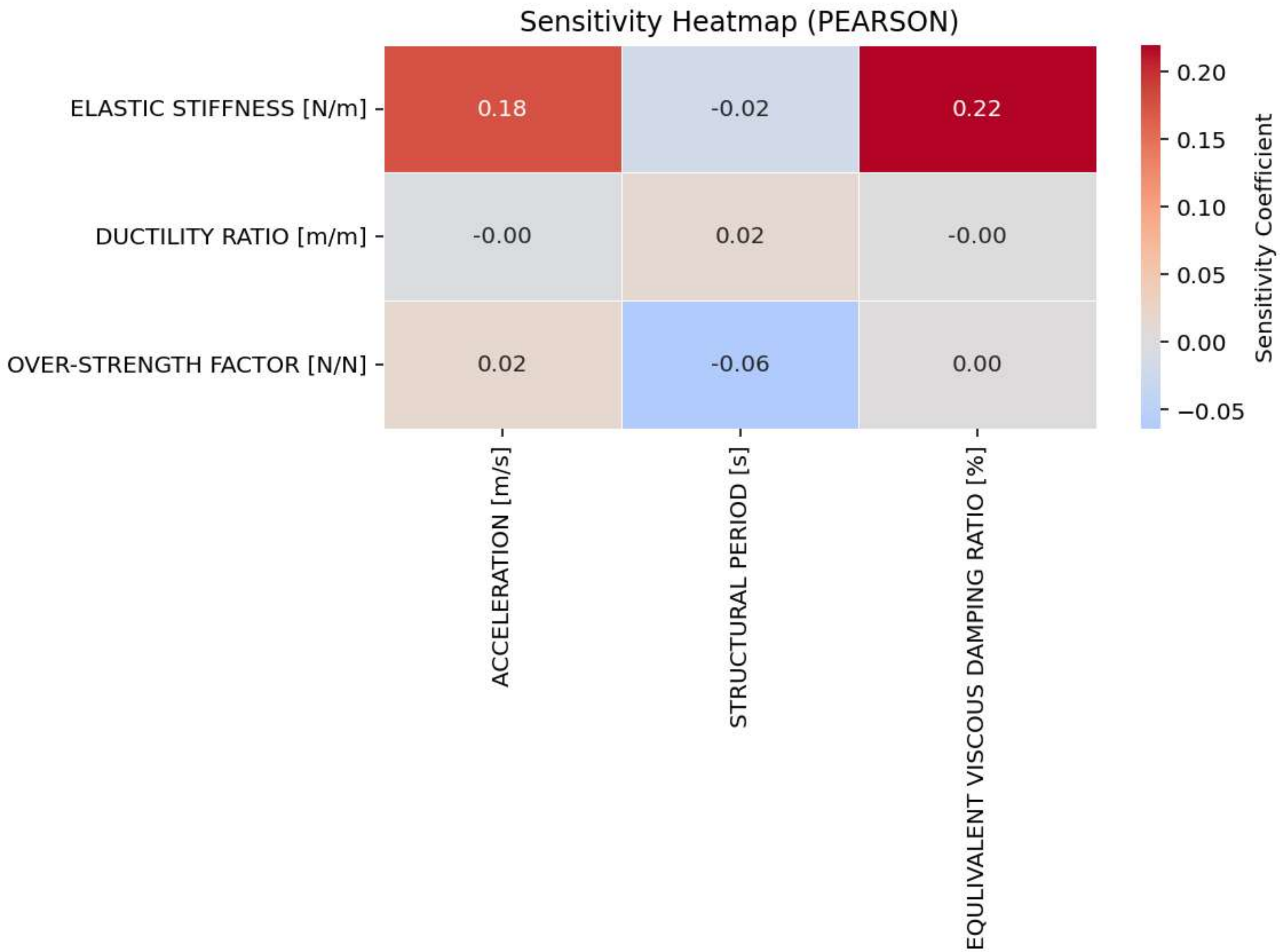
Histogram - Probability of Positive DUCTILITY DAMAGE INDEX FROM FREE-VIBRATION ANALYSIS is 0.00 %
Normal PDF
Quantile 0.25: 0.000000e+00
Median: 0.000000e+00
Quantile 0.75: 0.000000e+00
Mean: 0.000000e+00
Mean-Std: 0.000000e+00
Mean+Std: 0.000000e+00
Frequency
0
20
40
60
80
100
−0.4
−0.2
0.0
0.2
0.4
DUCTILITY DAMAGE INDEX FROM FREE-VIBRATION ANALYSIS


Boxplot of DUCTILITY DAMAGE INDEX FROM FREE-VIBRATION ANALYSIS
Mean: 0.000000e+00
Mean-Std: 0.000000e+00
Mean+Std: 0.000000e+00
Q2: 0.000000e+00
Q3: 0.000000e+00
Q1: 0.000000e+00
Data
1
−0.04
−0.02
0.00
0.02
0.04
DUCTILITY DAMAGE INDEX FROM FREE-VIBRATION ANALYSIS


Histogram - Probability of Positive EQULIVALENT VISCOUS DAMPING RATIO [%] FROM FREE-VIBRATION ANALYSIS is 100.00 %
Normal PDF
Quantile 0.25: 3.450777e+00
Median: 5.052038e+00
Quantile 0.75: 6.228995e+00
Mean: 4.700437e+00
Mean-Std: 2.929150e+00
Mean+Std: 6.471723e+00
Frequency
EQULIVALENT VISCOUS DAMPING RATIO [%] FROM FREE-VIBRATION ANALYSIS


Boxplot of EQULIVALENT VISCOUS DAMPING RATIO [%] FROM FREE-VIBRATION ANALYSIS
Mean: 4.700437e+00
Mean-Std: 2.929150e+00
Mean+Std: 6.471723e+00
Q1: 3.450777e+00
Q2: 5.052038e+00
Q3: 6.228995e+00
Data
1
1
2
3
4
5
6
7
EQULIVALENT VISCOUS DAMPING RATIO [%] FROM FREE-VIBRATION ANALYSIS


Correlation Heatmap
DISPLACEMENT
VELOCITY
ACCELERATION
REACTION
EQULIVALENT VISCOUS DAMPING RATIO
1.00
0.91
0.91
0.91
0.83
0.91
1.00
1.00
1.00
0.82
0.91
1.00
1.00
1.00
0.82
0.91
1.00
1.00
1.00
0.82
0.83
0.82
0.82
0.82
1.00
Variable
Variable
1.000
0.975
0.950
0.925
0.900
0.875
0.850
0.825


Sensitivity Heatmap (PEARSON)
ELASTIC STIFFNESS [N/m]
DUCTILITY RATIO [m/m]
OVER-STRENGTH FACTOR [N/N]
0.18
-0.02
0.22
-0.00
0.02
-0.00
0.02
-0.06
0.00
ACCELERATION [m/s]
STRUCTURAL PERIOD [s]
EQUIVALENT VISCOUS DAMPING RATIO [%]
0.20
0.15
0.10
0.05
0.00
−0.05
Sensitivity Coefficient